

Common Customer Engagement Components

Notify users of critical updates with App Alerts. Prevent legal and regulatory violations by making informed decisions with License Validation Checks. Manage complex business processes and guide users through their tasks with clear steps with multi-step workflows and related lists. Maximize customer engagement efficiency with Multi-Object Components and Dynamic Tree Views. Create Quick and Custom Actions for smoother workflows. Visualize Sales Data for effective customer interactions.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

App Alerts

Deliver in-app notifications to sales reps in their flow of work by using App Alerts. Create alerts and assign them to specific territories to make sure that notifications reach the right sales reps. With App Alerts, enhance in-app communication, drive user engagement, and avoid missing opportunities.

Dynamic Tree View Components for Life Sciences

Configure a Dynamic Tree View component to show related Life Sciences Cloud for Customer Engagement data in an expandable hierarchy. Users can easily explore related records in one place without going to multiple pages, boosting visibility and saving time. For example, show the account hierarchy within a health system, or visualize how healthcare professionals (HCPs) and healthcare organizations (HCOs) are aligned under specific territories.

License Management in Life Sciences

Help your company comply to legal and regulatory standards, and adhere to audit-tracking measures with flexible license validity evaluations. License management in Life Sciences is a critical process that makes sure that healthcare providers (HCPs) have the required license and authorization to receive pharmaceutical samples.

Multi-Object Components for Life Sciences

The Life Sciences Cloud for Customer Engagement multi-object components consolidate fields from several related objects onto a single record page, making it easier for users to view and manage data. Showing fields from multiple objects together helps users avoid going to multiple pages and simplifies the workflows for accounts, inquiries, and visits.

Provider Bulk Sync

Provider Bulk Sync helps organizations to effectively and efficiently manage their relationships with Healthcare Organizations (HCOs) and Healthcare Providers (HCPs). It allows organizations to integrate HCP and HCO data from IQVIA OneKey directly into the Life Sciences Cloud data model. This process helps organizations get verified HCP data, which helps users have the right information for their customer engagement plans and activities. The verified data can be used for territory planning and other processes.

Quick and Custom Action Management

To simplify user experiences in Salesforce and the Life Sciences for Customer Engagement mobile app, leverage quick actions and custom actions. These actions provide convenient access to essential information and common tasks, which streamlines user navigation and workflows.

Related Lists for Life Sciences

Records in Salesforce can include details and links to other related records. The Life Sciences Cloud for Customer Engagement related list offers more customization and filtering options than standard related lists. If you create a Life Sciences Cloud for Customer Engagement workflow, you can also control the actions that users can take on related records based on the parent record's status, user role, and other business rules. These related lists are supported both online and offline in the Life Sciences Cloud mobile app.

Reports and Dashboards Administration

Use reports and dashboards to deliver information to your Life Sciences Cloud mobile app users in the ways that work best for them.

Run Batch Jobs

Automate the processing of large record volumes by running Apex batch jobs from the Admin Console.

Trigger Handler Administration

View and manage trigger handlers in the Life Sciences Customer Engagement Admin Console. All trigger handlers are inactive by default, except the DCRHandler trigger handler.

Workflows for Life Sciences

Life Sciences Cloud for Customer Engagement workflows automate your business processes and guide users through their tasks with clear steps. Configure the permissions and actions for each step of the workflow based on the users' context and role. Implement custom validations to make sure that users follow business processes, and provide checklists to help users understand how to use the workflow. These workflows are supported both online and offline in the Life Sciences Cloud mobile app.

App Alerts

Deliver in-app notifications to sales reps in their flow of work by using App Alerts. Create alerts and assign them to specific territories to make sure that notifications reach the right sales reps. With App Alerts, enhance in-app communication, drive user engagement, and avoid missing opportunities.

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How App Alerts Work

Use App Alerts to create notifications, define how and when they appear, and deliver them to the correct sales reps. With App Alerts, sales reps can view all alerts in one place, take immediate action, and avoid missing opportunities.

Additional Message Field for App Alerts

Improve the standard look and presentation of alerts on the mobile app. Use the Additional Message field on the App Alert object for custom message bodies, categorized tags, and the actions that users can take directly from the alert.

Set Up App Alerts

Send relevant and targeted notifications, recommendations, and guided actions to sales reps on web and mobile apps. Configure the trigger handlers, assign access to the Apex classes, and activate the Life Sciences Cloud for Customer Engagement app for the standard user profile. Add the Lightning component to the record page to show Alerts to your sales reps.

Create an App Alert

App alerts are the broader, contextual notifications that appear in specific application areas. Choose a context type to control where the alert appears: All, Object, and Tab.

Record Alert Configuration for App Alerts

Target notifications associated with specific records, such as Accounts and Healthcare Providers, to your users. You can create a record alert manually or trigger it automatically using the Business Rules Engine.

Create App Alert Territories

App Alert Territories represents a junction between an Alert, either an App Alert or a Record Alert, and a Territory where the alert is sent. Use App Alert Territories to associate an alert with one or more territories so that only the users assigned to those territories see the message.

Mobile App Configuration for App Alerts

Configure the object schema for supported App Alerts objects and generate a metadata cache to make sure your sales reps receive timely alerts on the Life Sciences Cloud Mobile app.

Create a Metadata Cache

Package the object scheme configuration into a downloadable metadata cache that the mobile app can use. Create a metadata cache to make sure that the app consistently uses the latest metadata definitions, even when a network connection is unavailable.

How App Alerts Work

Use App Alerts to create notifications, define how and when they appear, and deliver them to the correct sales reps. With App Alerts, sales reps can view all alerts in one place, take immediate action, and avoid missing opportunities.

REQUIRED EDITIONS

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You can create App Alerts for sales reps in three ways, depending on your business needs.

- **Create Alerts Manually:** Create records for the Record Alert, App Alert, and App Alert Territories objects. To control who sees an alert, map it to one or more territories by using the App Alert Territories object, which helps you target alerts geographically or organizationally. Only the users assigned to the mapped territories see the alerts during the specified time period.
- **Create Alerts Automatically:** Eliminate the need for manual intervention by automating alerts to trigger based on specific conditions or user actions. For example, create rules in the Business Rules Engine or use Flows to send alerts based on real-time context or business rules automatically. See [Create Record Alerts Using Business Rules Engine](#).
- **Surface Alerts from External Systems:** If you have critical information stored outside of Salesforce, use this method to bring it in as an alert. See [Show External System Record Alerts in Salesforce](#).

Where and How Users See Alerts

Show the alerts directly in the Life Sciences Commercial app, so sales reps can stay informed in their flow of work.

On mobile, sales reps open alerts from the bell icon in the app header. The message pane organizes alerts into tabs for notifications, recommendations, and guided actions. Depending on the alert type, here are some actions users can perform on the mobile app:

- View alerts grouped by type in the message pane
- Dismiss or resolve alerts and if needed, provide a reason
- Complete tasks or actions directly from alerts
- Go to external websites, apps, or internal records

Web: Alerts are shown in the Notifications List Lightning web component on the Notifications and Recommendations tab. Add this component to a specific record page, the app's homepage, or an app page. It adapts based on the context and visibility rules that you configure. Here are the actions users can perform on the web app.

- Guided actions are grouped in the Notifications section
- Users can view alerts and mark them as Read

How Alert Visibility and Behavior Are Controlled

Control alert content and interaction by using the **AdditionalMessage** field on the App Alert object. This field supports a JSON structure for defining deep links, external URLs, action buttons, and tags. See [Additional Message in App Alerts](#).



Note If the alert includes actions, use the `actionName` field with values such as Dismissed and Resolved. You can also define required and optional reasons that users select when they dismiss the alert. These responses are stored in the App Alert User Response object.

Alert visibility is controlled by the **AlertType** and the **ValidUntilDate** fields.

- For mobile: Alerts remain visible until the user marks them as Read. If no `ValidUntilDate` value is set,

they stay visible indefinitely.

- For web: Notifications and recommendations are visible for up to 7 days from the effective date, unless dismissed or removed earlier. Guided Actions stay active until dismissed or resolved.

Here's how alert behavior and visibility work across platforms:

Feature	Web	Mobile
Sorting Logic	If multiple alerts appear on the same record, the most recent one, based on the effective date, appears at the top of the list.	Same as web: alerts are sorted by effective date, with the newest alert shown first.
Actions	Users can view and mark a message as Read.	Users can mark a message as Dismissed or Resolved. These actions are supported in the message pane. When you configure the alert to require a dismissal reason, users can provide one, as optional or required. The alert is then removed from the list, and the related alert status is updated via App Alert User Response.
Deep Links and Action Buttons	Not supported. Alerts can't include buttons or direct links to records or external systems.	Fully supported through the AdditionalMessage JSON field. You can include buttons, deep links, internal or external links, and tags in your alerts.

Additional Message Field for App Alerts

Improve the standard look and presentation of alerts on the mobile app. Use the Additional Message field on the App Alert object for custom message bodies, categorized tags, and the actions that users can take directly from the alert.

REQUIRED EDITIONS

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Additional Message configuration includes three main parts: Message, Tags, and Actions.

Message

Customize the alert message body by using plain text, hyperlinks, and deep links that direct users to specific records or tabs. Use deep links to guide users to targeted actions directly from the alert. The message supports:

- Basic HTML formatting, including <a> tags for hyperlinks
- Deep links to open tabs, records, and quick actions
- Web links to external content and documentation

Supported link formats for the Additional Message field.

The table outlines the link formats that you can use in the Additional Message field on the App Alerts object. These formats use HTML <a> tags to create deep links and external links. The table also explains the behavior and context of each link format.

Name	Examples	Description
Message Body	<pre>"message": "Agentforce just got smarter. Explore new features like guided visit prep, account insights, and action suggestions – all designed to help you plan faster and engage more effectively. Tap to see what's new and boost your daily workflow ."</pre>	
External Web Page	<pre>what's new and boost your daily workflow</pre>	• Opens the link inside the in-app browser when no target attribute is specified. • Opens the link inside the device's default browser, like Safari instead of the in-app browser.

Name	Examples	Description
	<pre>Search Today weather in Browser</pre>	
Tabs	<pre>Account Page</pre>	Opens a Salesforce tab directly, such as the Account tab.
Specific Record	<pre>YYYYY</pre>	Opens a specific record, such as Account, using its record ID.
Quick Actions	<pre>Plan with Dr. Jones</pre>	Triggers a predefined quick action on a specific record, such as logging a visit.
External App Deep Linking	<pre></pre>	Opens a screen or channel in a supported third-party app, such as Slack, using a valid deep link.



Note Line breaks aren't supported in the message body. Double quotes (") in link tags must be escaped using backslash (\").

Home Office Announcements

On App Alert records with the Announcement type, the Additional Message fields support these HTML tags so you can customize the app alerts that appear as notifications on the Life Sciences Commercial

home page.

- `<b>`
- `<i>`
- `<u>`
- `<s>`
- `<p>`
- `<br>`
- `<h1>`, `<h2>`, `<h3>`, `<h4>`, `<h5>`, `<h6>`

Tags

Use tags to categorize additional messages and visually highlight their context and purpose. These tags appear as labels in the alert that helps users quickly recognize the message. Each tag requires a title, a short label identifying the tag, for example, What's New.

Assign one or more tags to each message. Tags can represent campaign names, feature highlights, or categories such as Territory, Engagement, and Product Update.

This example demonstrates how tags are defined in JSON format.

```
{
  "tags": [ "Agentforce", "What's New", "Next Gen LSC" ]
}
```

Actions

Use actions to define what happens when a user interacts with a message. Configure actions to open a web URL, launch a deep link, or enable the user to dismiss or resolve the message. Use these options to direct users to relevant content or mark the notification as handled.

Supported Actions

For actions that involve opening a web URL or a deep link, each action must include at least a title and an action. For the dismiss or resolve action type, the required fields are title, actionable, and `actionName`; the `action` field is not necessary. In addition, there are parameters available to guide user responses when they dismiss or resolve actions.

This table lists the required fields for the two action types.

Action Type	Required Fields	JSON Examples
Open a Web URL or Deep Link	• Title: The label shown on the	<pre>{</pre>

Action Type	Required Fields	JSON Examples
	action button. Action: The URL or deep link that the action directs to.	<pre>"notificationActions": [{ "title": "Agentforce 2.0", "action": "https://ww w.theskyplanner.com/wh at-is-salesforce-einst ein/" }] }</pre>
Dismiss or Resolve Alerts	- Title: The label shown on the action button. - Actionable: Indicates whether the action is interactive or clickable for end users. When the actionable property is set to true, the action button becomes clickable, and the system records the user's response when the action is triggered. Skip this field for actions that open a web URL or deep link. - actionName: Specifies the outcome when users interact with the action. (optional) dismissOptions: A list of predefined options for the user to choose from when they dismiss the message. (optional) dismissOptionRequired: Indicates whether selecting a dismiss option is mandatory.	<pre>{ "notificationActions": [{ "title": "Dismiss", "actionable": true, "actionName": "Dismiss ed", "dismissOptions": ["Acknowledged", "Expired"], "dismissOptionRequire d": true }] }</pre>

See Also

[Create an App Alert](#)

Set Up App Alerts

Send relevant and targeted notifications, recommendations, and guided actions to sales reps on web and mobile apps. Configure the trigger handlers, assign access to the Apex classes, and activate the Life Sciences Cloud for Customer Engagement app for the standard user profile. Add the Lightning component to the record page to show Alerts to your sales reps.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

USER PERMISSIONS NEEDED

To configure App Alerts:	Life Sciences Commercial Admin
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- Profiles are configured, and permission sets are assigned to users based on their roles. See [Assign Life Sciences Cloud for Customer Engagement Permission Sets](#).
 - If you're using territory targeting, ensure your territory model includes territory types, active territory models, and user assignments to territories. For more information, see [Set Up Territory Types and Territories](#).
 - Required data is available for related objects, such as Accounts, App Alerts, Record Alerts, and App Alert Territories.
1. Make sure that your Salesforce org uses the default sharing settings because for App Alerts record sharing is based on territory assignments.
 - In the [sharing settings](#) for the App Alert and Record Alert objects, verify that default internal access is set to **Private**.
 - On each user profile, verify that the **View All Records** permission is disabled for the App Alert and Record Alert objects.
 2. Configure trigger handlers for App Alerts.
 - a. From the App Launcher, find and select **Admin Console**.
 - b. Click **Trigger Handler Administration**.
 - c. Turn on **NotificationTerritorySharingHandler**.
Check if a record on the LifeScienceTriggerHandler object with the developer name NotificationTerritorySharingHandler was created. If no record exists, create one.
 3. From Setup, assign access to the required Apex classes through the built-in Life Sciences Field Sales Representative or, optionally, through a custom profile created by you.
 - LWCSqlConnector
 - NotificationsComponentController
 - SystemController
 - TerritorySwitcherCmpController

See [Set Apex Class Access from Permission Sets](#) and [Set Apex Class Access from Profiles](#).

4. Add the built-in Notifications List Lightning component to the Life Sciences Cloud for Customer Engagement app.
Add the Lightning component on a record, home, or app page.
 - a. From the App Launcher, find and select **Life Sciences Commercial**.
 - b. Click the  icon, and then click **Edit Page**.
 - c. In the Components section, find and select **Notifications List** in the search box.
 - d. Drag the component to the canvas.
 - e. Save your changes.
5. If you selected Announcement as the type of alert, add the LSC Home Announcement Lightning component to the Life Sciences Cloud for Customer Engagement app.
Add the Lightning component on a record, home, or app page.
 - a. From the App Launcher, find and select **Life Sciences Commercial**.
 - b. Click the  icon, and then click **Edit Page**.
 - c. In the Components section, find and select **LSC Home Announcements**.
 - d. Drag the component to the canvas.
 - e. Save your changes.

See Also

[Create an App Alert](#)

[Record Alert Configuration for App Alerts](#)

[Create App Alert Territories](#)

Create an App Alert

App alerts are the broader, contextual notifications that appear in specific application areas. Choose a context type to control where the alert appears: All, Object, and Tab.

REQUIRED EDITIONS

Available in: Lightning Experience

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USER PERMISSIONS NEEDED

To configure App Alerts:	Life Sciences Commercial Admin
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1. From the App Launcher, find and select **App Alerts**.
2. Click **New**.
3. Enter a subject as needed.
4. Select the type of alert.
 - Announcement: For broad internal communication, such as home office announcements.



Note Use Salesforce Files to upload images and create image URLs for home office

announcements. Make sure to use HTTPS image URLs rather than HTTP, as HTTPS is more secure. See [Upload Images in your Salesforce Org](#). If images in home announcements don't appear, check whether the image URLs are from a domain added as a trusted URL. See [Manage Trusted URLs](#).

- Notification: For short, contextual updates to inform users about relevant actions or statuses.
 - Recommendation: To guide users toward the next best action or suggestion.
 - Guided Action: To direct users through specific tasks by using predefined instructions.
5. Select the display context type, as needed.
 - Object: Shows the alert on the record pages of a specific object. On web and mobile, if the object type is populated, the alert appears on all records of that object. You must populate the DisplayContextName with the object name.
 - Tab: Shows alerts on a specific tab in the app. The tab must be identified by the DisplayContextName value.
 - Global: Shows the alert across all records. Global is the default value applied if you leave the DisplayContextType field blank. On the web and mobile, global alerts appear on the homepage.
 6. Select the effective start date and time.
Indicates the date and time from when the alert becomes visible to users.
 7. If needed, select the effective end date and time.
 - On mobile, notifications and recommendations are visible by default for up to 7 days from the effective start date, unless removed earlier.
 - If no ValidUntilDate value is set for the web, alerts are visible indefinitely until the user marks them as Read.
 8. If needed, in Additional Message, enter the JSON object.
This field supports a JSON structure that enables deep links, external website links, action buttons, and tags. If the alert is actionable, it must be marked as Dismissed or Resolved through the actionName attribute in the JSON.
 9. If you want to prevent the alert from popping up on mobile devices, select **Silent**.
Users can still view messages in the message pane on mobile.
 10. Save your changes.

See Also

[Additional Message Field for App Alerts](#)
[Set Up App Alerts](#)

Record Alert Configuration for App Alerts

Target notifications associated with specific records, such as Accounts and Healthcare Providers, to your users. You can create a record alert manually or trigger it automatically using the Business Rules Engine.

REQUIRED EDITIONS

Available in: Lightning Experience

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Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

For more information on creating a record in the Record Alert object, see [Create a Record Alert](#).

To trigger alerts automatically, use Business Rules Engine to define the criteria to show a record alert when a record meets specified conditions. See [Create Record Alerts Using Business Rules Engine](#).

See Also

[Set Up App Alerts](#)

Create App Alert Territories

App Alert Territories represents a junction between an Alert, either an App Alert or a Record Alert, and a Territory where the alert is sent. Use App Alert Territories to associate an alert with one or more territories so that only the users assigned to those territories see the message.

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USER PERMISSIONS NEEDED

To configure App Alerts:	Life Sciences Commercial Admin
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1. From the App Launcher, find and select **App Alert Territories**.
2. Click **New**.
3. In Alert Reference Record, select the app alert record.
4. Find and select the territory that you want to link to the selected record.
5. Save your changes.



Note Remove all associated App Alert Territories before you delete the App Alert record.

See Also

[Set Up App Alerts](#)

Mobile App Configuration for App Alerts

Configure the object schema for supported App Alerts objects and generate a metadata cache to make sure your sales reps receive timely alerts on the Life Sciences Cloud Mobile app.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

USER PERMISSIONS NEEDED

To configure App Alerts:	Life Sciences Commercial Admin
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Here's the set of objects that require object schema configuration.

- AppAlert
 - AppAlertUserResponse
 - AppAlertTerritories
 - RecordAlert
1. From the App Launcher, find and select **Life Sciences Commercial**.
 2. Click **Admin Console**.
 3. On the Life Sciences Customer Engagement Setup page, click **Mobile**.
 4. On the navigation pane, click **Object Metadata Cache Configuration**.
 5. Click **New**.
 6. Enter the details.
 - a. Enter the name of the entity configuration.
 - b. If you want to activate the configuration, select **Is Active**.
 - c. Select the users that you want to assign the entity to, based on their profiles or permission sets.
 - d. If needed, select the attachment data for the object.
For example, select **Cache**.
 - e. If needed, in Delta Date field, select the applicable fields for the object.
 - f. To define a logic-based schema or data to store reference points, select the **Configuration** type.
 - g. To determine if the synchronization is unidirectional, select **One Way Sync**.
 - h. In Where Soql: select one of these.
 - Simple: To define conditions by using basic filters.
 - Advanced: To write a custom SOQL expression.
 - i. Select how you want the conditions to be evaluated.
 - All of the conditions are met: To apply the rule only when all conditions are true.
 - Any of the conditions are met: To apply the rule when at least one condition is true.
 - Customize the logic: To define manually how multiple conditions are evaluated.
 7. Save your changes.

See Also

[Customize the Life Sciences Cloud Mobile App](#)
[Generate Metadata Cache](#)

Create a Metadata Cache

Package the object scheme configuration into a downloadable metadata cache that the mobile app can use. Create a metadata cache to make sure that the app consistently uses the latest metadata definitions, even when a network connection is unavailable.

REQUIRED EDITIONS

Available in: Lightning Experience

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USER PERMISSIONS NEEDED

To configure App Alerts:	Life Sciences Commercial Admin
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 **Important** Run the metadata cache every time you configure the mobile app.

1. From the App Launcher, find and select **Admin Console**.
2. On the Modules Setup page, click the **Mobile** feature.
3. On the left navigation panel, click **Metadata Cache**.
4. On the Entities Setup page, click **New**.
5. Enter the details.
 - a. Enter the name for the metadata cache.
 - b. Select the type of mobile UI.
For example, select **Tab**.
 - c. Enter a name of the tab.
 - d. Enter a number to determine the sequence in which you want this tab to appear on the admin console.
 - e. Select the user profiles that can view this tab.
 - f. To activate the configuration, select **Is Active**.
 - g. To save your changes, click **Create New Cache**.

Dynamic Tree View Components for Life Sciences

Configure a Dynamic Tree View component to show related Life Sciences Cloud for Customer Engagement data in an expandable hierarchy. Users can easily explore related records in one place without going to multiple pages, boosting visibility and saving time. For example, show the account hierarchy within a health system, or visualize how healthcare professionals (HCPs) and healthcare organizations (HCOs) are aligned under specific territories.

To create a Dynamic Tree View component, configure the objects that appear in the component. Then, create the component that references the object configurations that you defined. Finally, add the component to a Lightning record page.

Create Object Configurations for a Dynamic Tree View Component

Configure the data that you want to show for each object in a Life Sciences Cloud for Customer Engagement Dynamic Tree View component.

Configure a Dynamic Tree View Component

Create a dynamic tree view component configuration to show Life Sciences Cloud for Customer Engagement data for related objects in a hierarchical view.

Add a Dynamic Tree View Component to a Record Page

Use Lightning App Builder to add your Dynamic Tree View component to the parent object's Lightning record page in Life Sciences Cloud for Customer Engagement.

Create Object Configurations for a Dynamic Tree View Component

Configure the data that you want to show for each object in a Life Sciences Cloud for Customer Engagement Dynamic Tree View component.

REQUIRED EDITIONS

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USER PERMISSIONS NEEDED

To create a dynamic tree view object configuration:	Life Sciences Commercial Admin permission set
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To create, edit, and delete field sets:	Customize Application
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Before you create a dynamic tree view object configuration, create field sets for each object that you want to show in the component. Field sets define the fields shown in the component itself and in an info icon on the component. Then, create the dynamic tree view object configurations.

1. From the App Launcher, find and select **Admin Console**, and then select **Component Library**.
2. Select **Dynamic Tree View Object Configurations**, and then create one.
3. Enter the basic details, such as the name for the dynamic tree view object configuration and whether it's active.
4. Enter the API name of the object that this configuration represents in the dynamic tree view component.
5. To filter the records shown in the dynamic tree view component, enter custom filter logic. For example, enter `Status = 'Active'` to show only the records in the Active state.
6. Choose whether to show percentage fields as a progress bar instead of a numeric value.
7. In Parent Object Lookup Field, enter the API name of the lookup field on this object that references the parent object.
8. In the Object Column Field Set and Object Info Field Set fields, enter the API names of the field sets

that you created for this object.

9. Choose whether to show column headers for this object's fields in the dynamic tree view component.
10. Enter the API names of the fields that you want to use for filtering or sorting records in the component, and define the sort order for the records shown.
11. Save your changes.
12. For each object that you want to show in the dynamic tree view component, create a similar configuration.

Next, create a dynamic tree view component configuration that references the object configurations that you created.

See Also

[Create and Edit Field Sets](#)

Configure a Dynamic Tree View Component

Create a dynamic tree view component configuration to show Life Sciences Cloud for Customer Engagement data for related objects in a hierarchical view.

REQUIRED EDITIONS

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USER PERMISSIONS NEEDED

To create a dynamic tree view component configuration:	Life Sciences Commercial Admin permission set
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1. From the App Launcher, find and select **Admin Console**, and then select **Component Library**.
2. Open the **Dynamic Tree View Component Configurations** page, and then click **New**.
3. Enter the basic details for the component, such as its name and whether it's active.
4. To filter the records shown for all objects in the component, enter a custom filter logic.
For example, enter *Name = 'AccountName'* to show the records related to a specific account.
5. In the Child Object Configuration, Parent Object Configuration, and Grandchild Object Configuration fields, enter the API names of the dynamic tree view object configurations that you created for each object you want to show in the component.
6. Choose whether to show a search bar and to let users filter records in the component.
7. Select whether the relationship between the parent and child objects uses a junction object. If it does, enter the details.
 - The API name of the junction object.
 - The API name of the lookup field on the junction object that references the grandparent object.
 - The API name of the lookup field on the junction object that references the related object.

8. Save your changes.

After you save your dynamic tree view component configuration, copy its API name. You need this information to add your component to the parent object's record page in Lightning App Builder.

Add a Dynamic Tree View Component to a Record Page

Use Lightning App Builder to add your Dynamic Tree View component to the parent object's Lightning record page in Life Sciences Cloud for Customer Engagement.

REQUIRED EDITIONS

Available in: Lightning Experience

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USER PERMISSIONS NEEDED

To create and save Lightning pages in Lightning App Builder:

Customize Application

To access the Dynamic Tree View component:

Life Sciences Commercial Admin permission set

1. From Setup, in the Quick Find box, enter *Lightning App Builder*, and then select it.
2. Edit a record page for the parent object, or create one.
3. Search the list of components for Dynamic Tree View, and then drag the component onto the page.
4. In the properties pane, enter the API name of the Dynamic Tree View Component Configuration that you created in Admin Console.
5. To add visibility rules based on the record field, device type, or other filters, click **Add Filter**.
An eye icon indicates that visibility rules are applied.
6. Save your changes to the record page, and then activate the page to share it with your users.

See Also

[Get Help for Lightning App Builder](#)

License Management in Life Sciences

Help your company comply to legal and regulatory standards, and adhere to audit-tracking measures with flexible license validity evaluations. License management in Life Sciences is a critical process that makes sure that healthcare providers (HCPs) have the required license and authorization to receive pharmaceutical samples.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

During sample disbursement, sales reps verify whether the healthcare provider (HCP) receiving the sample is licensed and eligible to practice in a given territory. For certain samples, users must also validate whether the HCP is licensed to prescribe, dispense, and administer controlled or dangerous drugs.

With License Management, you can configure custom formulas to evaluate the validity of an HCP's license. You can also configure the license validity icons on Lightning page, and in visits, searches, and lists.

Configure Custom Formulas for License Validations

Prepare your org for License Management by configuring custom validation rules for evaluating the status of HCP licenses.

Types of Licenses

License Management helps you configure license evaluations for these types of licenses: state license numbers (SLNs), state distributor licenses (SDLs), and drug enforcement administration (DEA) number. You can create custom validity formulas to control where these indicators are shown and meet your company's specific needs.

Configure Custom Formulas for License Validations

Prepare your org for License Management by configuring custom validation rules for evaluating the status of HCP licenses.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

USER PERMISSIONS NEEDED

To create custom formulas:	Life Sciences Commercial Admin permission set
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Create a custom formula field by using a checkbox as the formula return type. This checkbox serves as an automated check to indicate whether the license validation is successful or not.

1. From the object management settings for Business License, go to Fields & Relationships.
2. Select **New**.
3. In Data Type, select **Formula**, and then click **Next**.
4. Add a field label.
The field name is automatically populated.
5. In Formula Return Type, select **Checkbox** and click **Next**.
6. Add a formula for the validation evaluation.
For example, to check if a license is active, select Advanced Formula and then insert isActive.
7. To make sure your formula is valid, select **Check Syntax**, and then click **Next**.
8. Select the profiles the formula is visible to, and click **Next**.
The page layout is automatically selected. Keep it as is.
9. Save your changes.

See Also

[Build a Formula Field](#)

Types of Licenses

License Management helps you configure license evaluations for these types of licenses: state license numbers (SLNs), state distributor licenses (SDLs), and drug enforcement administration (DEA) number. You can create custom validity formulas to control where these indicators are shown and meet your company's specific needs.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

Life Sciences Cloud provides standard license validity formulas. With License Management, you can configure flexible custom formulas for validity checks. You can also select where the validity indicators are shown and which profiles can see it.

State License Numbers (SLNs)

SLNs are medical licenses for US healthcare providers (HCPs) and ensure that the HCPs are authorized to practice medicine and receive or prescribe samples within a specific state. SLNs are typically checked before sales reps disburse samples to them.

State Distributor Licenses (SDLs)

SDLs are issued to entities or facilities distributing dangerous drugs in certain states. In certain states, entities that distribute dangerous drugs need specific licenses, such as Terminal Distributor of Dangerous

Drugs (TDDD) in Ohio.

Drug Enforcement Administration (DEA) Number

The DEA number is a license number issued by the Drug Enforcement Administration to providers who prescribe controlled substances. DEA numbers are important for compliance and audit purposes, and must be validated when samples include controlled substances.

Define State License Number Validations

Check the medical license issued to an HCP to practice within a US state by configuring state license number validations (SLNs). This configuration lets sales reps check a healthcare provider's license validity before sharing product samples.

Define State Distributor License Validations

To help sales reps comply with regulatory standards while distributing samples, create state distributor license (SDL) validations. This configuration ensures that the person or the facility that distributes high-risk products, such as those categorized as dangerous drugs, has the necessary license to do so.

Define Drug Enforcement Administration License Validations

Help your org comply with federal rules by configuring Drug Enforcement Administration (DEA)-issued identifiers for HCPs who prescribe, dispense, and administer controlled substances. Your sales reps can confirm an HCP's license before providing samples. The DEA number ensures thorough regulation, tracking, and audit of controlled substances.

Define State License Number Validations

Check the medical license issued to an HCP to practice within a US state by configuring state license number validations (SLNs). This configuration lets sales reps check a healthcare provider's license validity before sharing product samples.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

USER PERMISSIONS NEEDED

To create license validations:

Life Sciences Commercial Admin permission set



Note Make sure you've created a custom formula field on the Business License object.

1. From the App Launcher, find and select **Admin Console**.
2. Select **License Management**.
3. In the left navigation pane, select **State License Number Setup**.

4. Make these changes.
 - a. Choose whether this setting applies to the org or to a specific profile.
 - b. In the License Validity Check section, specify the validation formula by selecting the custom formula that you configured in the Business License Object.
 - c. To change the formula configured in the Business License object, select **Modify Formula**.
 - d. Choose whether you want to evaluate the license validity and show the validity icon in visits, searches and lists, and Lightning pages.
 - e. Save your changes.

Define State Distributor License Validations

To help sales reps comply with regulatory standards while distributing samples, create state distributor license (SDL) validations. This configuration ensures that the person or the facility that distributes high-risk products, such as those categorized as dangerous drugs, has the necessary license to do so.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

USER PERMISSIONS NEEDED

To create license validations:	Life Sciences Commercial Admin permission set
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Note Make sure you've created a custom formula field on the Business License object.

1. From the App Launcher, find and select **Admin Console**.
2. Select **License Management**.
3. In the left navigation pane, select **State Distributor License Setup**.
4. Make these changes.
 - a. Choose whether this setting applies to the org or to a specific profile.
 - b. In the License Validity Check section, specify the validation formula by selecting the custom formula that you configured in the Business License Object.
 - c. To change the formula configured on the Business License object, select **Modify Formula**.
 - d. Select **Evaluate license validity and show icon in visits**.
 - e. Move the states that require the license validation evaluation to the Selected Values list.
 - f. Save your changes.

Define Drug Enforcement Administration License Validations

Help your org comply with federal rules by configuring Drug Enforcement Administration (DEA)-issued identifiers for HCPs who prescribe, dispense, and administer controlled substances. Your sales reps can confirm an HCP's license before providing samples. The DEA number ensures thorough regulation,

tracking, and audit of controlled substances.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

USER PERMISSIONS NEEDED

To create license validations:

Life Sciences Commercial Admin permission set



Note Make sure you've created a custom formula field on the Business License object.

1. From the App Launcher, find and select **Admin Console**.
2. Select **License Management**.
3. In the left navigation pane, select **DEA Number Setup**.
4. Make these changes.
 - a. Choose whether this setting applies to the org or to a specific profile.
 - b. In the License Validity Check section, specify the validation formula by selecting the custom formula that you configured in the Business License Object.
 - c. To change the formula configured on the Business License object, select **Modify Formula**.
 - d. Select **Evaluate license validity and show icon in visits**.
 - e. Save your changes.

Multi-Object Components for Life Sciences

The Life Sciences Cloud for Customer Engagement multi-object components consolidate fields from several related objects onto a single record page, making it easier for users to view and manage data. Showing fields from multiple objects together helps users avoid going to multiple pages and simplifies the workflows for accounts, inquiries, and visits.

How Multi-Object Components Work

With the multi-object Lightning components included with Life Sciences Customer Engagement, users can create, edit, and view records that include fields from multiple objects on a single page.

Set Up the Multi-Object Create and Edit Components

Assign overrides for standard Salesforce buttons to help users in Life Sciences Cloud for Customer Engagement create or edit records with fields from multiple objects.

Set Up the Multi-Object Record View Component

To help users in Life Sciences Cloud for Customer Engagement view fields from multiple objects on a single record page, add the Multi-Object Record View component to record pages.

How Multi-Object Components Work

With the multi-object Lightning components included with Life Sciences Customer Engagement, users can create, edit, and view records that include fields from multiple objects on a single page.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

Multi-Object Lightning Components

The Life Sciences Cloud for Customer Engagement managed package includes these Lightning components that support multi-object layouts.

- The MultiEntityCreateOverride component overrides the New action so users can populate fields and create multiple records at once.
- The MultiEntityEditOverride component overrides the Edit action so users can update fields from multiple records at once.
- The Multi-Object Record View component shows the details from related objects on a single Lightning record page.

Supported Objects

The account object supports these components to include fields from the healthcare provider object. To override the New button on the account object, use the Search Before Create component. See [Configure Search Before Account Creation](#).

- MultiEntityEditOverride
- Multi-Object Record View

The inquiry object supports these components to include fields from the case object.

- MultiEntityCreateOverride
- MultiEntityEditOverride
- Multi-Object Record View

The visit object supports these components to include fields from the provider visit object. These components are also supported on the provider visit object to show visit fields.

- MultiEntityEditOverride
- Multi-Object Record View

Considerations

When you set up multi-object components, keep these considerations in mind.

- Multi-object components are available only for supported objects.
- Multi-object components support objects with multiple object types, picklist assignments, and picklist field dependencies.
- Inline editing isn't supported on record detail pages that include a multi-object component.
- When users create records with a multi-object component, if one record has an error, no new records are created.
- When users create or edit records using a multi-object component, validation rules and errors appear at the top of the page rather than inline.
- The details that appear in each component depend on the existing parent records and related records.

Set Up the Multi-Object Create and Edit Components

Assign overrides for standard Salesforce buttons to help users in Life Sciences Cloud for Customer Engagement create or edit records with fields from multiple objects.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

USER PERMISSIONS NEEDED

To access and work with Life Sciences Cloud for Customer Engagement multi-object components: Life Sciences Commercial Admin permission set

To override standard buttons: Customize Application

Multi-object Lightning components are supported only for these Life Sciences objects.

Lightning Component	Objects
Isc4ce:MultiEntityCreateOverride	Inquiry
Isc4ce:MultiEntityEditOverride	• Account • Inquiry • Provider Visit • Visit

1. From the management settings for the object whose buttons you want to override, go to Button, Links, and Actions.
2. In the row for the New action, select **Edit** from the dropdown menu.
3. For the Lightning Experience Override, select **Lightning component**, and then select **Isc4ce:MultiEntityCreateOverride** from the dropdown.
4. Save your changes.
5. Similarly, assign the override for the Edit action. Select **Isc4ce:MultiEntityEditOverride** as the override for Lightning Experience.

The overrides that you apply to Lightning Experience also apply to the Life Sciences Cloud mobile app.

Set Up the Multi-Object Record View Component

To help users in Life Sciences Cloud for Customer Engagement view fields from multiple objects on a single record page, add the Multi-Object Record View component to record pages.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

USER PERMISSIONS NEEDED

To access and work with Life Sciences Cloud for Customer Engagement multi-object components:	Life Sciences Commercial Admin permission set
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To create and save Lightning pages in Lightning App Builder:	Customize Application
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When you install the Life Sciences Cloud for Customer Engagement managed package, the Multi-Object Record View component is added to the default Lightning record pages for these objects.

- Account
- Inquiry
- Provider Visit
- Visit

You can add the component to other Lightning record pages for the Account and Inquiry objects and object types.

1. From Setup, in the Quick Find box, enter *Lightning App Builder* and select it.
2. Edit a record page, or create one.
3. Search the list of components for Multi-Object, and then drag the Multi-Object Record View component onto the page's Details tab.

4. Remove any other detail components from the tab.
5. Save your changes to the record page, and then activate the page to share it with your users.

See Also

[Get Help for Lightning App Builder](#)

Provider Bulk Sync

Provider Bulk Sync helps organizations to effectively and efficiently manage their relationships with Healthcare Organizations (HCOs) and Healthcare Providers (HCPs). It allows organizations to integrate HCP and HCO data from IQVIA OneKey directly into the Life Sciences Cloud data model. This process helps organizations get verified HCP data, which helps users have the right information for their customer engagement plans and activities. The verified data can be used for territory planning and other processes.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

[Set Up Provider Bulk Sync by Using Built-In MuleSoft Integration](#)

Integrate Life Sciences Cloud with IQVIA's OneKey to import healthcare provider and organization information from an external system into Salesforce Life Sciences Cloud for Customer Engagement. Life Sciences Cloud offers a prebuilt suite of integrations apps, such as the Provider Bulk Sync API and the Onekey System API. These apps are deployed by using MuleSoft Direct to seamlessly integrate with and import healthcare provider information from various external data sources.

[Create a Copy of Healthcare Provider Address Records by Using Flows and Batch Jobs](#)

Regulatory restrictions in certain countries prevent the retrieval of healthcare provider (HCP) addresses via IQVIA OneKey. To ensure data completeness, use Salesforce batch jobs and flows to trigger invocable actions that create copies of address records from the main affiliated healthcare organizations (HCOs) for all corresponding HCP records. The Process Affiliation Address Records invocable action is triggered when a healthcare professional's affiliation status changes, such as joining or leaving a healthcare organization. The Process Contact Point Address Records invocable action is triggered when a healthcare organization's address is added or updated.

Set Up Provider Bulk Sync by Using Built-In MuleSoft Integration

Integrate Life Sciences Cloud with IQVIA's OneKey to import healthcare provider and organization information from an external system into Salesforce Life Sciences Cloud for Customer Engagement. Life Sciences Cloud offers a prebuilt suite of integrations apps, such as the Provider Bulk Sync API and the Onekey System API. These apps are deployed by using MuleSoft Direct to seamlessly integrate with and import healthcare provider information from various external data sources.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

Integrate MuleSoft Direct

To address end-to-end business needs, integrate Salesforce apps with external systems by using Integration Solutions with MuleSoft Direct. To set up MuleSoft Direct, see [Integrate Solutions with Mulesoft Direct](#). To set up integration for provider bulk sync, see [Explore MuleSoft Direct Integration Apps](#).

After the MuleSoft Direct integration is complete, you can sync healthcare provider and organization data. After a sync job is initiated, you can get the status of the sync job and abort the sync job. To ensure your provider data stays up to date, schedule a bulk data synchronization job from Salesforce to automatically pull information from external systems like OneKey regularly.



Note If healthcare provider (HCP) address details aren't synced, it's likely due to regulatory restrictions in that country. Alternatively, you can use Salesforce Flows and batch jobs that trigger invocable actions. This approach allows you to copy the missing address information directly from the affiliated healthcare organizations (HCOs). See [Create a Copy of Healthcare Provider Address Records by Using Flows and Batch Jobs](#)

See Also

[Explore MuleSoft Direct Integration Apps](#)
[Provider Bulk Sync Intergration App](#)
[Provider Bulk Sync API](#)

Create a Copy of Healthcare Provider Address Records by Using Flows and Batch Jobs

Regulatory restrictions in certain countries prevent the retrieval of healthcare provider (HCP) addresses via IQVIA OneKey. To ensure data completeness, use Salesforce batch jobs and flows to trigger invocable actions that create copies of address records from the main affiliated healthcare organizations (HCOs) for all corresponding HCP records. The Process Affiliation Address Records invocable action is triggered when a healthcare professional's affiliation status changes, such as joining or leaving a healthcare organization. The Process Contact Point Address Records invocable action is triggered when a healthcare organization's address is added or updated.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

- **Process Contact Point Address:** This invocable action synchronizes HCP address information with the affiliated HCOs. When an HCO address is created, the action creates a copy of the address for all affiliated HCPs. When an HCO address becomes inactive, the action deactivates the corresponding copied addresses for all affiliated HCPs.
- **Process Affiliation Address:** This invocable action ensures HCPs always have accurate address information based on their active affiliations with healthcare organizations (HCOs). When an HCP is affiliated with an HCO, the action copies the HCO addresses and links them to the HCP record. When the affiliation ends, the action deactivates the copied addresses to maintain up-to-date data.

To copy HCP address records from affiliated HCOs, create the required batch jobs and Salesforce Flows. The batch job triggers the flow, which in turn invokes the invocable action to create the address records.

Create Flows to Trigger Invocable Actions to Copy Addresses

Create Salesforce Flows that are used by batch jobs to trigger invocable actions that create a copy of healthcare provider (HCP) address records from the affiliated healthcare organization (HCOs).

Create Batch Jobs to Copy Healthcare Provider Addresses

Create Batch Management jobs that trigger the necessary flows, which in turn call the invocable actions to create a copy of the Contact Point Addresses and Process Provider Affiliation Addresses.

Sequence Address Batch Jobs by Using a Flow

To make sure that address synchronization batch jobs run in the correct order, create a flow that automatically initiates the Process Provider Affiliation Addresses batch job only after the Process HCP Contact Point Addresses batch job completes.

Create Flows to Trigger Invocable Actions to Copy Addresses

Create Salesforce Flows that are used by batch jobs to trigger invocable actions that create a copy of healthcare provider (HCP) address records from the affiliated healthcare organization (HCOs).

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

USER PERMISSIONS NEEDED

To create a flow:	Manage Flow
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Create a Flow to Copy HCP Contact Point Addresses

Create the flow to trigger the Process Contact Point Address invocable action.

1. From Setup, enter *Flows* in the Quick Find box, select **Flows**, and then select **New Flow**.
2. In the New Flow window, select **Autolaunched Flow (No Trigger)**.
3. Create variables for input values.
 - a. From the toolbox, select **Manager** and then select **New Resource**.
 - b. For Resource Type, select **Variable**.
 - c. For API Name, enter *cpaRecordId*.
 - d. For Data Type, select **Text**.
 - e. For Description, enter *The ID of the contact point address that was passed into the flow.*
 - f. Select **Available for input**.
 - g. Click **Done**.
4. Create an action element that calls the invocable action and uses the *cpaRecordId* variable.
 - a. For Category, select **Ls Commercial**.
 - b. Select the Process Contact Point Address Records invocable action.
 - c. For Label, enter *Process Contact Point Address Records*.
The API name is automatically populated.
 - d. Set *cpaRecordId* as an input value.
 - e. Click **Done**.
5. Save and activate the flow.

Create a Flow to Copy Provider Affiliation Addresses

Create the flow to trigger the Process Affiliation Address invocable action.

1. From Setup, enter *Flows* in the Quick Find box, select **Flows**, and then select **New Flow**.
2. In the New Flow window, select **Autolaunched Flow (No Trigger)**.
3. Create variables for input values.
 - a. From the toolbox, select **Manager** and then select **New Resource**.
 - b. For Resource Type, select **Variable**.
 - c. For API Name, enter *affiliationRecordId*.
 - d. For Data Type, select **Text**.
 - e. For Description, enter *the ID of the provider affiliation that was passed into the flow.*
 - f. Select **Available for input**.
 - g. Click **Done**.
4. Create an action element that calls the invocable action and uses the *affiliationRecordId* variable.
 - a. For Category, select **Ls Commercial**.
 - b. Select the Process Provider Affiliation Address Records invocable action.
 - c. For Label, enter *Process Affiliation Address Records*.
The API name is automatically populated.

- d. Set `affiliationRecordId` as an input value.
 - e. Click **Done**.
5. Save and activate the flow.

Create Batch Jobs to Copy Healthcare Provider Addresses

Create Batch Management jobs that trigger the necessary flows, which in turn call the invocable actions to create a copy of the Contact Point Addresses and Process Provider Affiliation Addresses.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

USER PERMISSIONS NEEDED

To create a batch job: System Administrator profile

Create a Batch Job to Copy HCP Contact Point Addresses

Create the batch job to invoke the flow that triggers the Process Contact Point Address invocable action.

1. From Setup, in the Quick Find box, enter *Batch Management*, and select it.
2. Click **New**.
3. Complete the required fields.
 - a. For Name, enter *Process HCP Contact Point Addresses*.
 - b. For Process Type, select **Flow**.
 - c. For Execution Process, select the Process HCP Contact Point Addresses flow.
 - d. For Group, enter *Life Sciences*.
 - e. Enter the batch size, retry count, and retry interval.
 - f. Click **Next**.
4. Select the Flow Input Variable as **cpaRecordId** and Object as **Contact Point Address**.
5. Enter the conditions shown in the table.

Field	Operator	Type	Value
Last Modified Date	Greater Than Or Equal To	Input Variable	startTime
Last Modified Date	Less Than Or Equal To	Input Variable	endTime
Last Modified By ID	Equals	Value	Integration User
Created Date	Greater Than Or Equal	Input Variable	startTime

Field	Operator	Type	Value
	To		
Created Date	Less Than Or Equal To	Input Variable	endTime
Active	Equals	Value	False

6. Save and activate the batch job.

Create Batch Job to Copy Provider Affiliation Addresses

Create the batch job to invoke the flow that triggers the Process Affiliation Address Records invocable action.

1. From Setup, in the Quick Find box, enter *Batch Management*, and select it.
2. Click **New**.
3. Complete the required fields.
 - a. For Name, enter *Process Provider Affiliation Addresses*.
 - b. For Process Type, select **Flow**.
 - c. For Execution Process, select the Process Provider Affiliation Addresses flow.
 - d. For Group, enter *Life Sciences*.
 - e. Enter the batch size, retry count, and retry interval.
 - f. Click **Next**.
4. Select the Flow Input Variable as **affiliationRecordId** and Object as **Provider Affiliation**.
5. Enter the conditions shown in the table.

Field	Operator	Type	Value
Last Modified Date	Greater Than Or Equal To	Input Variable	startTime
Last Modified Date	Less Than Or Equal To	Input Variable	endTime
Last Modified By ID	Equals	Value	Integration User
Affiliation Type	Equals	Value	Hard
Account Person Account	Equals	Value	True
Related Account Person Account	Equals	Value	False
Created Date	Greater Than Or Equal To	Input Variable	startTime
Created Date	Less Than Or Equal To	Input Variable	endTime
Active	Equals	Value	False

6. Save and activate the batch job.

Sequence Address Batch Jobs by Using a Flow

To make sure that address synchronization batch jobs run in the correct order, create a flow that automatically initiates the Process Provider Affiliation Addresses batch job only after the Process HCP Contact Point Addresses batch job completes.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

USER PERMISSIONS NEEDED

To create a flow: Manage Flow

1. From Setup, enter *Flows* in the Quick Find box, select **Flows**, and then select **New Flow**.
2. In the New Flow window, select **Platform Event - Triggered Flow**.
3. For Platform event, select **Batch Job Status Changed Event** and in Advanced settings, select **Run this flow as the user that triggered the event**.
4. Create a decision element that uses a record or answers from a screen input to determine a flow path.
 - a. On the flow canvas, add a Decision element.
 - b. Enter a label and API name for the element, for example *Is HCP Contact Point Addresses Batch Job Complete?*
 - c. Give labels for the outcomes and enter these conditions.

Resource	Operator	Value
Triggering BatchJobStatusChangedEvent -> Batch Job	Equals	Process_HCP_Contact_Point_Addresses's BatchJobDefinitionId Example batch job definition ID: 0maxx0000000001AAA
Triggering BatchJobStatusChangedEvent -> Batch Job	Does not Equal	Cancelled

- d. Click **Done**.
5. Create variables that store the batch job runtime parameters and make the value available for output.
 - a. From the toolbox, select **Manager** and then select **New Resource**.

- b. For Resource Type, select **Variable**.
 - c. For API Name, enter *batchJobRuntimeParameters*.
 - d. For Data Type, select **Text**.
 - e. Select **Available for output**.
 - f. Click **Done**.
6. Create a Get Records element to retrieve the HCP Contact Point Addresses batch job details.
 - a. On the flow canvas, add a Get Records element.
 - b. Enter a label and API name for the element, for example *Get Batch Job Record*.
 - c. For Object, select **Batch Job**.
 - d. Under Filter Account Records Condition Requirements, select **All Conditions are Met (AND)**.
 - e. For Field, enter *Id*, then select **Run ID**.
 - f. For Operator, select **Equals**.
 - g. For Value, select **Resource Triggering BatchJobStatusChangedEvent -> Batch Job**.
 - h. For How Many Records to Store, select **Only the first record**.
 - i. For How to Store Record Data, select **Choose fields and assign variables (advanced)**.
 - j. For Where to Store Field Values, select **In separate variables**.
 - k. For Field, select **Runtime Parameter**.
 - l. For Value, select **batchJobRuntimeParameters**.
 - m. Click **Done**.
7. Create these formulas to get the individual batch job runtime parameters.

API Name	Data Type	Formula
extractedStartTime	TEXT	<pre> LEFT (MID ({!batchJobRuntimeParameters}, FIND ('{"name":"startTime","value":"' , {!batchJobRuntimeParameters}) + LEN ('{"name":"startTime","value":"' , 50), FIND ('"', MID ({!batchJobRuntimeParameters}, FIND ('{"name":"startTime","value":"' , {!batchJobRuntimeParameters} </pre>

API Name	Data Type	Formula
		<pre>eters)) + LEN('{ "name": "startTime", "value": "" }, 50)) - 1)</pre>
extractedEndTime	TEXT	<pre>LEFT(MID({!batchJobRuntimeParameters}, FIND('{ "name": "endTime", "value": "" }, {!batchJobRuntimeParameters}) + LEN('{ "name": "endTime", "value": "" }, 50), FIND('"', MID({!batchJobRuntimeParameters}, FIND('{ "name": "endTime", "value": "" }, {!batchJobRuntimeParameters}) + LEN('{ "name": "endTime", "value": "" }, 50)) - 1)</pre>
startTimeAsDateTime	Date/Time	<pre>DATETIMEVALUE(SUBSTITUTE(SUBSTITUTE({!extractedStartTime}, "T", "</pre>

API Name	Data Type	Formula
		<code>) , "Z", ""))</code>
endTimeAsDateTime	Date/Time	<code>DATETIMEVALUE (SUBSTITUTE (SUBSTITUTE ({!extractedEndTime}, "T", ""), "Z", ""))</code>

8. Create an Action element to trigger the Process Provider Affiliation Addresses Batch job. This element uses the `startTimeAsDateTime` and `endTimeAsDateTime` formulas.
 - a. Select the Category as **Batch Job**, and then select the **Process Provider Affiliation Addresses** batch job.
 - b. For Label, enter *Trigger Process Provider Affiliation Address Batch Job*.
The API name is automatically populated.
 - c. In the Set input values section, select `startTime` as **startTimeAsDateTime** and `endTime` as **endTimeAsDateTime**.
 - d. Click **Done**.
9. Save and activate the flow.

Quick and Custom Action Management

To simplify user experiences in Salesforce and the Life Sciences for Customer Engagement mobile app, leverage quick actions and custom actions. These actions provide convenient access to essential information and common tasks, which streamlines user navigation and workflows.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

Quick actions provide direct access to core functionalities and help users initiate key processes without extensive searching. For example, from an account profile, users can schedule a visit, send an email, or start a remote visit from a visit record. This immediate accessibility reduces clicks and saves time.

Custom actions invoke Lightning components, flows, Visualforce pages, or canvas apps with the functionality that you define. Use custom actions to simplify commonly performed processes that are unique to your business needs.

Set Up and Manage Quick Actions

Define who can use specific Life Sciences Customer Engagement quick actions and where the actions appear. Make sure that the correct users see the correct options exactly when they need them.

REQUIRED EDITIONS

USER PERMISSIONS NEEDED	
To manage quick actions:	Life Sciences Commercial Admin permission set
	AND
	Customize Application

1. Create an action.
 - a. From the App Launcher, find and select **Life Sciences Commercial**, and then select **Admin Console**.
 - b. Select **Quick and Custom Action Administration**, and then select **Quick Actions**.
 - c. Select **New**.
 - d. Enter a label, name, action name, location, and sort order.
 - e. If needed, set optional values such as mobile only, sort order, icon, color, associated object, and record type.
 - f. Select the profiles and permission sets that can access the action.
 - g. Save your changes.
2. Manage existing actions.
 - a. From the App Launcher, find and select **LifeSciences Commercial**, and then select **Admin Console**.
 - b. Select **Quick and Custom Action Administration**, and then select **Quick Actions**.
 - c. From the dropdown, select the action.
 - Edit: Change the location, sort order, icon, color, associated object, and record type. Or, update the profiles and permission sets that have access to the action.
 - Activate/Deactivate: Turn the action on or off.
 - View Assignments: See a list of profiles and permission sets assigned to the action.
 - d. Save your changes.

Create Custom Actions

Invoke Lightning components, flows, Visualforce pages, or canvas apps with the functionality that you define for Life Sciences Customer Engagement users.

REQUIRED EDITIONS

USER PERMISSIONS NEEDED

To create custom actions:

Life Sciences Commercial Admin permission set

AND

Customize Application

1. From the App Launcher, find and select **Life Sciences Commercial**, and then select **Admin Console**.
2. Select **Quick and Custom Action Administration**, and then select **Custom Actions**.
3. Select **New**.
4. Enter a label and name for the custom action.
5. To define where the action is available to users, select an entity type.

Home Page	The action is available on the mobile app Home page.
SObject	The action is available on the object defined in the Entity Name field.
Stage Path	The action is available for stage paths.
Visit	The action is available to support custom visit management tasks.

6. To define where the action takes place, select the action type.

App	The action takes place in the mobile app.
URL	The action opens a new URL.
Utterance	The action sends a predefined message to Agentforce.

7. Select the target type.
 - Internal
 - External
8. Enter values in the additional fields to meet your business needs.
9. Save your work.

See Also

[Quick Actions](#)

Related Lists for Life Sciences

Records in Salesforce can include details and links to other related records. The Life Sciences Cloud for Customer Engagement related list offers more customization and filtering options than standard related lists. If you create a Life Sciences Cloud for Customer Engagement workflow, you can also control the actions that users can take on related records based on the parent record's status, user role, and other business rules. These related lists are supported both online and offline in the Life Sciences Cloud mobile app.

When you add the Customer Engagement related list to a record page, you can:

- Customize the list's header, icon, and columns.
- Filter the records in the list.
- Populate fields on new records based on the parent record's data.
- Specify the record types that users can create.

Get Your Org Ready for Life Sciences Related Lists

Before you create a Life Sciences Customer Engagement related list, complete these steps.

Create Related Lists for Life Sciences

Add a Life Sciences Cloud for Customer Engagement related list to the page layout for Life Sciences objects, standard Salesforce objects, and custom objects. The related list appears in Life Sciences Cloud on desktop and in the mobile app.

Get Your Org Ready for Life Sciences Related Lists

Before you create a Life Sciences Customer Engagement related list, complete these steps.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

USER PERMISSIONS NEEDED

To configure the Life Sciences Customer Engagement related list:

Life Sciences Commercial Admin permission set

To create field sets and custom fields:

Customize Application

1. If needed, create a relationship between the parent and child object.
 - a. [Create a lookup field](#) from the child object to the parent object.
 - b. Add the field to the [child object's page layout](#).
2. To define the columns for the related list, [create a field set](#).

Add up to 10 fields, and arrange them in the order that you want the columns to appear in.

3. To grant users access to the related list child object and any fields used to filter the records in the list, [update permission sets](#).

These permissions make sure that users can create, read, edit, or delete records in the related list.

4. To further control when users can create, edit, and delete related records based on their role and the parent object record's status, [create a Life Sciences Customer Engagement workflow](#) for the parent object.
5. If needed, [create custom Lightning web components](#) for the New, Edit, and Delete actions, or other actions that you want to include in the related list.

You can also create a custom component for the View All link on related lists on iOS.

6. Set up the configurations for the related list to work in the Life Sciences Cloud mobile app.
 - a. [Set up object metadata cache configurations](#) for the related list objects.
 - b. [Generate a metadata cache](#) to package the object schema configuration that the Life Sciences Cloud mobile app uses for online and offline access.

Create Related Lists for Life Sciences

Add a Life Sciences Cloud for Customer Engagement related list to the page layout for Life Sciences objects, standard Salesforce objects, and custom objects. The related list appears in Life Sciences Cloud on desktop and in the mobile app.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

USER PERMISSIONS NEEDED

To configure the Life Sciences Cloud for Customer Engagement related list:	Life Sciences Commercial Admin permission set
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To create and save Lightning pages in Lightning App Builder:	Customize Application
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When you install the Life Sciences Cloud for Customer Engagement managed package, some related lists are automatically added to these default Lightning record pages.

- Account
- Inquiry
- Visit

You can add a Customer Engagement related list to other standard and custom Lightning record pages.

1. From Setup, in the Quick Find box, enter *Lightning App Builder* and select it.

2. Edit a record page, or create one.
3. Search the list of components for Related List, and then drag the Related List – Life Sciences component onto the record page's Related tab.
4. In the properties pane, customize the list.
 - a. Enter the API name of the child object whose related records you want to show.
 - b. Enter the API name of the field set that you created on the child object.
The field set defines the fields you want to use as the list's columns.
 - c. Enter the title of the related list.
 - d. Enter the name of the Salesforce Lightning Design System (SLDS) icon that you want to include in the list header. Use the format **group:name**.
 - e. Enter the API name of the action handler component that handles the edit and delete actions.
To use the standard Edit and Delete actions for each record in the related list, enter *StandardUpdateDeleteHandler*.
 - f. To filter the records in the list, enter a WHERE clause in SOQL format.
For example, enter *AccountId = 'recordId'*. You can also filter on a polymorphic field. For example, *ContactPointReferenceId IN (SELECT ID from ContactPointAddress WHERE ParentId = !recordId)*.
 - g. Enter the API name of the action handler component that handles the new action.
To show the standard New action so users can create related records from the list, enter *StandardNewAction*.
 - h. If the child object has record types, specify the record types that users can create from the list. Enter the record type API names in a semicolon separated list.
If you don't specify record types, users can create all types.
 - i. To simplify data entry, enter the API name of the relationship field to be populated automatically on new related records. The relationship field must be added to the child record's page layout.
 - j. To open the standard View All page, enter *StandardViewAll* in the View All Component field. To show users a custom View All page on iOS devices, enter the name of the Lightning web component that you created.
 - k. Choose whether to show the number of records in the list header.
5. To add visibility rules based on the record field, device type, or other filters, click **Add Filter**.
The eye icon on the component indicates that visibility rules are applied.
6. Save your changes to the record page, and then activate the page to share it with your users.

After you add the related list to the page layout, create a metadata cache for the required profiles so users can access the data in the mobile app when they are offline.

See Also

[Get Help for Lightning App Builder](#)

Reports and Dashboards Administration

Use reports and dashboards to deliver information to your Life Sciences Cloud mobile app users in the ways that work best for them.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

Manage Reports and Dashboards Settings

Define settings for the reports and dashboards shown to users on the Life Sciences Cloud mobile app.

REQUIRED EDITIONS

USER PERMISSIONS NEEDED

To manage reports and dashboards settings:

Life Sciences Commercial Admin permission set

AND

Customize Application

1. From the App Launcher, find and select **Life Sciences Commercial**, and then select **Admin Console**.
2. Select **Reports and Dashboards**, and then select **Reports and Dashboards Settings**.
3. Select the mobile report settings based on your business needs.
 - Show Sales Data Chart
 - Show reports
4. Select the refresh frequency.
5. Save the settings.

Limitations for Charts and Dashboards in the Life Sciences Cloud Mobile App

Learn about the limitations for report charts and dashboards in the Life Sciences Cloud mobile app. Not all chart types render correctly or are supported.

After you set up reports and dashboards, you can add them to Lightning page components to help users visualize data from the Customer Engagement home page and record detail pages. In Lightning App Builder, you can add these standard components to Lightning pages.

- Dashboard: Supported on the home page
- Report Chart: Supported on the home page and Lightning record pages

You can also show reports or dashboards in separate tabs within the Life Sciences Cloud mobile app.

Limitations for Bar and Column Charts

Standard bar charts and column charts are replaced with stacked charts in the Life Sciences Cloud mobile app.

Unsupported Chart Types

These chart types aren't supported in the Life Sciences Cloud mobile app

- Funnel charts
- Scatter charts

See Also

[Customize Reports and Dashboards](#)
[Reports and Dashboards](#)
[Get Help for Lightning App Builder](#)

Run Batch Jobs

Automate the processing of large record volumes by running Apex batch jobs from the Admin Console.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

USER PERMISSIONS NEEDED

To run batch jobs:	Life Sciences Commercial Admin permission set
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1. From the App Launcher, find and select **Life Sciences Commercial**, and then select **Admin Console**.
2. Select a tile and find the batch job you want to run. For example, Products.
3. To run the batch job now, click **Run Now**.
 - a. Select a territory and its subordinates from the appropriate level in the territory hierarchy.
 - b. Select the batch size for the job, drag the slider handle. The default value is 200.
 - c. Click **Run**.
4. To schedule the job for a later time, click **Schedule**.
 - a. Select a frequency.
 - b. To select the batch size of the job, drag the slider handle. The default value is 200.
 - c. Save your changes.

Trigger Handler Administration

View and manage trigger handlers in the Life Sciences Customer Engagement Admin Console. All trigger handlers are inactive by default, except the DCRHandler trigger handler.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

Manage Trigger Handlers

View all Life Sciences Customer Engagement trigger handlers in one place, and turn them on or off.

Trigger Handlers for the Account - AssessmentTask Objects

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the Account, AccountPlan, AccountPlanObjective, AccountPlanParticipant, ActionPlan, ActivityPlan, ActivityPlanTerritory, AppAlertTerritory, and AssessmentTask objects.

Trigger Handlers for the Case - GoalAssignment Objects

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the Case, CommSubConsentCmplSnpsht, CommSubscription, CommSubscriptionConsent, ContactPointAddress, ContactPointEmail, ContactPointPhone, ContactPointSocial, ContentDocument, ContentDocumentLink, Expense, ExpenseParticipant, and GoalAssignment objects.

Trigger Handlers for the HealthcareProvider - PresentationLinkedPage Objects

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the HealthcareProvider, HealthcareProviderNpi, HealthcareProviderSpecialty, InventoryCntProdtBatchItem, InventoryCountAssessment, InventoryOperation, LifeScienceEmail, LifeSciMarketableProduct, Location, Presentation, PresentationForum, and PresentationLinkedPage objects.

Trigger Handlers for the MedicalInsight - Subject Objects

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the MedicalInsight, UserReaction, Subject and SubjectAssignment objects.

Trigger Handlers for the ProductBatchItem - ProductTransfer Objects

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ProductBatchItem, ProductDisbursement, ProductGuidance, ProductionBatch, ProductRequest, ProductRequestLineItem, ProductTerrDtIAvailability, ProductTerritoryAvailability, and ProductTransfer objects.

Trigger Handlers for the ProviderAcctProductInfo - PrvdVstSmpLmtTransaction Objects

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ProviderAcctProductInfo, ProviderAcctTerritoryInfo, ProviderActivityGoal, ProviderAffiliation, ProviderVisit, ProviderVisitChangedEvent, ProviderVisitDtIProductMsg, ProviderVisitMarketingItem, ProviderVisitProdDetailing, ProviderVisitProdDiscussion, ProviderVisitRqstSample, and PrvdVstSmpLmtTransaction objects.

Trigger Handlers for the SurveySubject - Visitor Objects

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the SurveySubject, Territory2, TerritoryAcctRcmdAction, TerritoryBusinessPlan, TerritoryContentTplAsgnt, TerritoryUserDowntime, Visit, and Visitor objects.

Manage Trigger Handlers

View all Life Sciences Customer Engagement trigger handlers in one place, and turn them on or off.

REQUIRED EDITIONS

USER PERMISSIONS NEEDED	
To manage trigger handlers:	Life Sciences Commercial Admin permission set
	AND
	Customize Application

Before you can use the Admin Console to manage trigger handlers, you must enable it.

1. From the App Launcher, find and select **Life Sciences Commercial**, and then select **Admin Console**.
2. Select **Trigger Handler Administration**.
3. Find specific trigger handlers by using the Object API Name dropdown or Search field.
4. Turn the trigger handler on or off.

Trigger Handlers for the Account - AssessmentTask Objects

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the Account, AccountPlan, AccountPlanObjective, AccountPlanParticipant, ActionPlan, ActivityPlan, ActivityPlanTerritory, AppAlertTerritory, and AssessmentTask objects.

REQUIRED EDITIONS

Available in: Lightning Experience
Available in: Enterprise and Unlimited Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

You can also review the trigger handlers that apply to all objects.

Account

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the

Account object.

Trigger Name	Description	Trigger condition
NewAccountSampleLimitInitHandlerforHCO	Initializes the HCO accounts for the Provider Sample Limit Template.	afterInsert
NewAccountSampleLimitInitHandler	Initializes the HCP accounts for the Provider Sample Limit Template.	afterInsert

AccountPlan

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the AccountPlan object.

Trigger Name	Description	Trigger condition
AccountPlanDeleteChildrenHandler	Handles the deletion of Account Plan Children records.	beforeDelete
AccountPlanChangeStatusHandler	Updates the completion percentage of the Account Plan records based on updates in the related Account Plan Objective records.	beforeInsert, beforeUpdate

AccountPlanObjective

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the AccountPlanObjective object.

Trigger Name	Description	Trigger condition
SetAccPlanPercentCompleteHandler	Updates the completion percentage of the Account Plan Objective records based on updates in the related Action Plan records.	beforeInsert, afterInsert, beforeUpdate, afterUpdate, afterDelete

AccountPlanParticipant

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the AccountPlanParticipant object.

Trigger Name	Description	Trigger condition
AccountPlanToParticipantSharingHandler	Shares Account Plan and hierarchical records with participants.	afterInsert, afterUpdate, afterDelete

ActionPlan

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ActionPlan object.

Trigger Name	Description	Trigger condition
SetAccObjectPercentCompleteHandler	Updates the completion percentage of the Action Plan records to 100% when the parent Account Plan's status is changed to Completed.	beforeInsert, afterInsert, beforeUpdate, afterUpdate, afterDelete
ActionPlanDeleteChildrenHandler	Deletes all the action plans related to a parent account plan when the account plan is deleted.	beforeDelete, afterDelete
SetSprintPercentCompleteHandler	Updates the completion percentage of the sprint based on the average completion percentages of the related action plans.	afterInsert, afterUpdate, afterDelete
SetGoalAsgmtPercentCompleteHandler	Updates the Completion Percentage of Goal Assignment records.	beforeInsert, afterInsert, beforeUpdate, afterUpdate, afterDelete

ActivityPlan

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ActivityPlan object.

Trigger Name	Description	Trigger condition
ActivityPlanTerritoryValidationHandler	Validates the alignment of territories with activity plans and displays an error message if the conditions aren't met.	beforeInsert, afterInsert, beforeUpdate, afterUpdate, beforeDelete, afterDelete, afterUndelete
ActivityPlanValidationHandler	Identifies and prevents duplicate activity plan records and displays an error message if the conditions aren't met.	beforeInsert, afterInsert, beforeUpdate, afterUpdate, beforeDelete, afterDelete, afterUndelete
PAPATerritoryUpdateHandler	Adjusts the activity plan goals and creates the corresponding sharing record for the adjusted territory.	beforeInsert, beforeUpdate, afterInsert
PAPAApproveProcessHandler	Prevents users from creating more than the allowed number of provider activity plan adjustment requests.	beforeInsert, beforeUpdate

ActivityPlanTerritory

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ActivityPlanTerritory object.

Trigger Name	Description	Trigger condition
ActivityPlanTerritoryModHandler	Validates and prevents an update of the Territory field in an Activity Plan Territory record or deletion of the Activity Plan Territory record if the associated activity plan is active or the status of the activity plan is not In Progress. If either condition is met, an error is displayed on the record page, preventing the update or deletion.	beforeInsert, afterInsert, beforeUpdate, afterUpdate, beforeDelete, afterDelete, afterUndelete
ActivityPlanTerritorySharingHandler	Manages the creation of	beforeInsert, afterInsert,

Trigger Name	Description	Trigger condition
	sharing records upon territory alignment and supports the update and deletion of sharing records to reflect territory re-alignment or removal.	beforeUpdate, afterUpdate, beforeDelete, afterDelete, afterUndelete

ALL

Review the names, descriptions, and trigger conditions of the trigger handlers associated with all objects.

Trigger Name	Description	Trigger condition
StagePathPermissionsHandler	Enforces permissions defined in Life Sciences workflows. Validates whether users can create, edit, or delete records or work with related files at the current stage in the workflow path. Shows an error message if users don't have permission to perform actions at that workflow stage.	afterInsert, beforeUpdate, beforeDelete
DCRHandler	Triggers changes to Account and related objects.	beforeInsert, afterInsert, beforeUpdate, afterUpdate, afterDelete

AppAlertTerritory

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the AppAlertTerritory object.

Trigger Name	Description	Trigger condition
NotificationTerritorySharingHandler	Creates sharing records once a territory has been assigned to an alert.	afterInsert, afterUpdate

AssessmentTask

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the AssessmentTask object.

Trigger Name	Description	Trigger condition
SetActionPlanPercentCompleteHandler	Updates the completion percentage of Action Plan records.	afterInsert, afterUpdate
AssessmentTaskDeleteAPIItemHandler	Deletes all the assessment tasks related to an action plan when the action plan is deleted.	beforeDelete
AssessmentTaskSharingHandler	Shares Assessment Task records with assignees.	afterInsert, afterUpdate

Trigger Handlers for the Case - GoalAssignment Objects

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the Case, CommSubConsentCmplSnpsht, CommSubscription, CommSubscriptionConsent, ContactPointAddress, ContactPointEmail, ContactPointPhone, ContactPointSocial, ContentDocument, ContentDocumentLink, Expense, ExpenseParticipant, and GoalAssignment objects.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

You can also review the trigger handlers that apply to all objects.

Case

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the Case object.

Trigger Name	Description	Trigger condition
CaseMSLQueueHandler	Monitors and manages case status changes. Handles the medical inquiry queue of the	beforeUpdate

Trigger Name	Description	Trigger condition
	medical science liaison users before a case record is updated. When an inquiry is submitted and the status changes to Submitted, the trigger handler assigns the case to the appropriate MSL queue. When the status changes to Assigned, the trigger handler reassigns ownership to the current user for handling the inquiry.	
CaseChangeOwnershipHandler	Handles the sharing of the medical inquiry records with the sales representatives after a case record is updated. When a case transitions to Submitted or Responded status, the handler creates CaseShare records to give read access to the case creator or owner. This process maintains data transparency in inquiry management workflows by ensuring that the original creator or owner retains visibility into the case, even if ownership changes.	afterUpdate
CaseShareToCreatorHandler	Handles the change of ownership of the medical inquiries between the sales representative and the medical science liaison users after a case record is updated. When a case's OwnerId changes, the ownership change automatically cascades to all related inquiry objects, such as InquiryQuestions and SubjectAssignments. The	afterUpdate

Trigger Name	Description	Trigger condition
	handler queries for cases with updated ownership, retrieves the related records, and updates the Owner on these child objects, ensuring data consistency.	

CommSubConsentCmplSnpsht

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the CommSubConsentCmplSnpsht object.

Trigger Name	Description	Trigger condition
ConsentGiverSnapshotHandler	Validates if the consent giver is valid.	beforeInsert
ConsentSnapshotSharingHandler	Shares the consent snapshot with territories that are aligned with the account.	afterInsert

CommSubscription

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the CommSubscription object.

Trigger Name	Description	Trigger condition
ConsentNameCommSubHandler	Updates Communication Subscription Consent records when related Communication Subscription record names are changed.	beforeUpdate

CommSubscriptionConsent

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the CommSubscriptionConsent object.

Trigger Name	Description	Trigger condition
ConsentGiverConsentHandler	Validates if the consent-giver	beforeInsert, beforeUpdate

Trigger Name	Description	Trigger condition
	account exists and it's the same for the snapshot record.	
ConsentSharingHandler	Shares consent with territories that are aligned with the account.	afterInsert
ConsentUniqueRecordHandler	Handles changes to CommSubscriptionConsent and maintains unique records.	beforeInsert

ContactPointAddress

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ContactPointAddress object.

Trigger Name	Description	Trigger condition
AccountAddressPrimaryHandler	Ensures that each account maintains only one primary contact point address. It gets triggered whenever a contact point address record is created or updated, guaranteeing that the one primary rule is always enforced.	beforeInsert, afterInsert, beforeUpdate, afterUpdate
RelatedContactPointAddressHandler	Validates contact point address hierarchy rules to ensure child addresses are associated with person accounts and parent addresses are associated with business accounts.	beforeInsert, beforeUpdate

ContactPointEmail

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ContactPointEmail object.

Trigger Name	Description	Trigger condition
ContactPointEmailPopulateHandler	Updates healthcare provider records to maintain an accurate list of associated email addresses whenever contact point email records are created, modified, removed, or restored. This ensures the EmailIds field on the healthcare provider record reflects the current related email addresses and avoids unnecessary processing.	afterInsert, afterUpdate, afterDelete, afterUndelete

ContactPointPhone

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ContactPointPhone object.

Trigger Name	Description	Trigger condition
ContactPointPhoneHandler	Updates healthcare provider records to maintain an accurate list of associated phone numbers whenever contact point phone records are created, modified, removed, or restored. This ensures the PhoneNumbers field on the healthcare provider record reflects the current related telephone numbers and prevents redundant processing.	afterInsert, afterUpdate, afterDelete, afterUndelete

ContactPointSocial

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ContactPointSocial object.

Trigger Name	Description	Trigger condition
ContactPointSocialPopulateHandler	Updates healthcare provider records to maintain an accurate list of associated social media handles whenever contact point social records are created, modified, removed, or restored. This ensures the SocialIdentifiers field on the healthcare provider record reflects the current related social handles and prevents unnecessary execution.	afterInsert, afterUpdate, afterDelete, afterUndelete

ContentDocument

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ContentDocument object.

Trigger Name	Description	Trigger condition
ConsentContentDocumentHandler	Handles changes to ContentDocument.	afterInsert, afterUpdate, afterDelete, afterUndelete
ContentDocumentLockHandler	Prevents delete operation to Documents linked to Submitted Visits or related entities.	beforeUpdate, beforeDelete

ContentDocumentLink

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ContentDocumentLink object.

Trigger Name	Description	Trigger condition
ConsentContentDocumentLinkHandler	Handles changes to ContentDocumentLink.	beforeInsert, beforeDelete
ContentDocumentLinkLockHandler	Prevents insert/delete operation to Document Links for Submitted Visits or related entities.	beforeInsert, beforeDelete

Expense

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the Expense object.

Trigger Name	Description	Trigger condition
ExpenseLockHandler	Prevents insert/update/delete operation on Expense if it is locked.	beforeInsert, beforeUpdate, beforeDelete

ExpenseParticipant

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ExpenseParticipant object.

Trigger Name	Description	Trigger condition
ExpenseParticipantNameHandler	Stores the Account/Contact Participant name as the Expense Participant Name.	beforeInsert
ExpenseParticipantLockHandler	Prevents insert/update/delete operation on ExpenseParticipant/Allocations if it is locked.	beforeInsert, beforeUpdate, beforeDelete

GoalAssignment

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the GoalAssignment object.

Trigger Name	Description	Trigger condition
GoalAssignmentDeleteHandler	Deletes Goal Assignment records.	beforeDelete
SetTerrBusPlanPrctCompleteHandler	Updates the Completion Percentage of Territory Business Plan records.	beforeInsert, afterInsert, beforeUpdate, afterUpdate, afterDelete

Trigger Handlers for the HealthcareProvider - PresentationLinkedPage Objects

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the HealthcareProvider, HealthcareProviderNpi, HealthcareProviderSpecialty, InventoryCntProdtBatchItem,

InventoryCountAssessment, InventoryOperation, LifeScienceEmail, LifeSciMarketableProduct, Location, Presentation, PresentationForum, and PresentationLinkedPage objects.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

You can also review the trigger handlers that apply to all objects.

HealthcareProvider

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the HealthcareProvider object.

Trigger Name	Description	Trigger condition
HealthcareProviderHandler	Manages updates to healthcare provider records after creation, restoration, or modification. This trigger fetches and updates related email addresses, phone numbers, social identifiers, specialties, and NPI from their respective related records, while also preventing self-calling updates.	beforeInsert, beforeUpdate
HealthcareProviderAffiliationHandler	Creates a primary affiliation between a parent organization and an account on the Healthcare Provider object. If an existing affiliation is found, the handler will just set or unset the primary flag on that record. The role for the new affiliation defaults to the selected value from the active picklist on the provider affiliation; if no	afterInsert, afterUpdate

Trigger Name	Description	Trigger condition
	default role is selected, the handler throws a custom error.	

HealthcareProviderNpi

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the HealthcareProviderNpi object.

Trigger Name	Description	Trigger condition
HealthcareProviderNpiHandler	Updates healthcare provider records to maintain the correct National Provider Identifier (NPI) whenever healthcare provider NPI records are created, modified, removed, or restored. This ensures the NationalProviderIdentifier field on the healthcare provider record reflects the current associated NPI and avoids unnecessary execution.	beforeInsert, afterInsert, beforeUpdate, afterUpdate, afterDelete, afterUndelete

HealthcareProviderSpecialty

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the HealthcareProviderSpecialty object.

Trigger Name	Description	Trigger condition
HealthcareProviderSpecialtyHandler	Updates healthcare provider records to maintain an accurate list of associated specialty types whenever healthcare provider specialty records are created, modified, removed, or restored. This ensures the Specialties field on the	beforeInsert, afterInsert, beforeUpdate, afterUpdate, afterDelete, afterUndelete

Trigger Name	Description	Trigger condition
	healthcare provider record reflects the current related specialties and avoids redundant processing.	

InventoryCntProdBatchItem

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the InventoryCntProdBatchItem object.

Trigger Name	Description	Trigger condition
InventoryCountPBILockHandler	Prevents modifications to Inventory Count Product Batch Item records when the associated inventory count assessment is locked.	beforeUpdate, beforeDelete
InventoryCntPBIVValidationHandler	Prevents update/delete operation on InventoryCntProdBatchItem 's if unresolved Product disbursement records are present for the same product.	beforeInsert, beforeUpdate

InventoryCountAssessment

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the InventoryCountAssessment object.

Trigger Name	Description	Trigger condition
InventoryCountAssessmentLockHandler	Prevents modifications to Inventory Count Assessment records when they're in a locked state, and prevents the creation of duplicate "Initial" type inventory count assessments.	beforeInsert, afterInsert, beforeUpdate, beforeDelete

InventoryOperation

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the InventoryOperation object.

Trigger Name	Description	Trigger condition
InventoryOperationLockHandler	Prevents modifications to Inventory Operation records when they're in a locked state.	afterInsert, beforeUpdate, beforeDelete
InventoryOperationSyncTxnHandler	Prevents the creation of Return, Adjustment, Transfer In, and Transfer Out type operations when the inventory operation is in its final state, and checks for pending sync transactions before updating the Inventory Operation records.	beforeInsert, beforeUpdate

LifeScienceEmail

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the LifeScienceEmail object.

Trigger Name	Description	Trigger condition
EmailQueueImmediateHandler	Handles emails that are sent immediately instead of from a batch job.	afterInsert, afterUpdate
EmailQueueRecipientHandler	Populates additional information for email recipients.	beforeInsert, beforeUpdate
EmailQueueSenderHandler	Populates additional information for the email sender.	beforeInsert
EmailAttachmentHandler	Populates email attachments.	beforeInsert
SyncEmailResponseStatusToObjectHandler	Updates the invitation status on related records based on the email response status.	afterUpdate

LifeSciMarketableProduct

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the LifeSciMarketableProduct object.

Trigger Name	Description	Trigger condition
LifeSciMarketableProductMetadataHandler	Creates or updates the ProductMetadata field value in the Life Science Marketable Product object.	afterInsert, afterUpdate, beforeDelete, afterDelete

Location

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the Location object.

Trigger Name	Description	Trigger condition
LocationUserDupHandler	Prevents duplicate primary user assignments to inventory locations.	beforeInsert, beforeUpdate
LocationUserSharingHandler	Shares records with the primary user assigned to an inventory location.	beforeInsert, beforeUpdate, afterUpdate

Presentation

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the Presentation object.

Trigger Name	Description	Trigger condition
PresentationSharingHandler	Recalculates how Presentation Page records are shared after a presentation is deleted.	beforeDelete

PresentationForum

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the PresentationForum object.

Trigger Name	Description	Trigger condition
PresentationForumLockHandler	Prevents creating, editing, or deleting Presentation Forum records if the related presentation is locked.	beforeInsert, beforeUpdate, beforeDelete

PresentationLinkedPage

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the PresentationLinkedPage object.

Trigger Name	Description	Trigger condition
PresentationLinkedPageSharingHandler	Recalculates how Presentation Page records are shared after Presentation Linked Pages are added or removed.	beforeInsert, beforeDelete

Trigger Handlers for the MedicalInsight - Subject Objects

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the MedicalInsight, UserReaction, Subject and SubjectAssignment objects.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

You can also review the trigger handlers that apply to all objects.

MedicalInsight

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the MedicalInsight object.

Trigger Name	Description	Trigger condition
PublishMedicalInsightEventHandler	Publishes MedicalInsightEvent platform events whenever a	afterInsert, afterUpdate, afterDelete

Trigger Name	Description	Trigger condition
	medical insight record is created, updated, or deleted. This facilitates real-time event streaming and integration with external systems.	
RecalculateParentLinkedInsightsHandler	Updates the rollup summary fields on parent medical insight records when child insights are modified. This includes updating the ChildMedicalInsightCount, UpvoteCount, and LastUpvotedDate fields to make sure roll-up summaries are accurate.	afterInsert, afterUpdate, afterDelete
MedicalInsightSharingHandler	Automatically shares medical insight records with peers having the same territory and profile as the creator upon creation. This handler creates sharing records with edit access to make sure that peers can collaborate on insights.	afterInsert
InsightCascadeDeleteTriggerHandler	Automatically deletes child medical insight records when a top-level parent record is deleted. This cascade deletion prevents orphan records and keeps the insight hierarchy clean.	beforeDelete

Subject

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the Subject object.

Trigger Name	Description	Trigger condition
TopicValidationHandler	Prevents the creation of	beforeUpdate

Trigger Name	Description	Trigger condition
	circular parent-child relationships when updating Subject records for medical insights. This makes sure a valid single-parent hierarchy structure.	

SubjectAssignment

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the SubjectAssignment object.

Trigger Name	Description	Trigger condition
RecalculateInsightTopicNameHandler	Updates the TopicNames field on the related medical insight record whenever a Subject Assignment is created or deleted. This makes sure the processed insight accurately reflects the set of topics currently assigned to the record.	afterInsert, afterDelete

UserReaction

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the UserReaction object.

Trigger Name	Description	Trigger condition
UserReactionMedicalInsightHandler	Updates the UpvoteCount field on Medical Insight records when User Reaction records are created or deleted. It also deletes duplicate records to prevent multiple upvotes from the same user on the same insight.	afterInsert, afterDelete

Trigger Handlers for the ProductBatchItem - ProductTransfer Objects

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ProductBatchItem, ProductDisbursement, ProductGuidance, ProductionBatch, ProductRequest, ProductRequestLineItem, ProductTerrDtlAvailability, ProductTerritoryAvailability, and ProductTransfer objects.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

ProductBatchItem

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ProductBatchItem object.

Trigger Name	Description	Trigger condition
ProductBatchItemOwnershipHandler	Shares records in product batch item to sales rep in ProductBatchItem.AssigneeId .	afterInsert

ProductDisbursement

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ProductDisbursement object.

Trigger Name	Description	Trigger condition
ProductDisbursementInventoryOpCreation	Creates a corresponding Inventory Operation record when a disbursement is made.	afterInsert, afterUpdate
ProductDisbursementSubmitLockHandler	Prevents modifications to Product Disbursement records when the related Visit record is in a locked state, that is, when the visit is signed or submitted.	beforeInsert, beforeUpdate, beforeDelete

ProductGuidance

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ProductGuidance object.

Trigger Name	Description	Trigger condition
ProductGuidanceDeactivationHandler	Removes sharing if the Product Guidance record was deactivated.	afterUpdate

ProductionBatch

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ProductionBatch object.

Trigger Name	Description	Trigger condition
ProductBatchItemUpdateByLotHandler	Shares Product Batch Item records with the assignee or the current user if the Assignee ID is not specified.	afterUpdate

ProductRequest

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ProductRequest object.

Trigger Name	Description	Trigger condition
ProductRequestLockHandler	Prevents modifications to Product Request records when they're in a locked state.	beforeUpdate, beforeDelete

ProductRequestLineItem

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ProductRequestLineItem object.

Trigger Name	Description	Trigger condition
ProductRequestLineItemLockHandler	Updates the ordered quantity in the Territory	beforeInsert, beforeUpdate, beforeDelete

Trigger Name	Description	Trigger condition
	Product Quantity Allocation record when a product request line item is modified.	
PrdReqLineItmUpdateAllocationHandler	Prevents modifications to Product Request Line Item records when the associated product request is locked.	afterInsert, afterUpdate

ProductTerrDtlAvailability

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ProductTerrDtlAvailability object.

Trigger Name	Description	Trigger condition
ProductTerrDtlAvailabilityHandler	On trigger of these events the Product Guidance records shared with Territory are deleted.	afterUpdate, afterDelete

ProductTerritoryAvailability

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ProductTerritoryAvailability object.

Trigger Name	Description	Trigger condition
ProductTerritoryAvailabilityExclHandler	Deletes ProductGuidance share records when an Exclusion rule is created in ProductTerritoryAvailability.	afterInsert, afterUpdate

ProductTransfer

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ProductTransfer object.

Trigger Name	Description	Trigger condition
ProductTransferLockHandler	Prevents modifications to Product Transfer records	beforeUpdate, beforeDelete

Trigger Name	Description	Trigger condition
	when the associated inventory operation is locked.	
ProductTransferValidationHandler	Prevents invalid Transfer In and Transfer Out operations by checking transfer directions, and source and destination locations.	beforeInsert, beforeUpdate
ReceivedProductAllocationHandler	Updates Territory Product Quantity Allocation records when a product transfer is received.	afterInsert, afterUpdate
TransferOutTriggerHandler	Creates a Transfer In record when a user receives a product transfer due to another user submitting a Transfer Out record.	afterInsert, afterUpdate

Trigger Handlers for the ProviderAcctProductInfo - PrvdVstSmplLmtTransaction Objects

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ProviderAcctProductInfo, ProviderAcctTerritoryInfo, ProviderActivityGoal, ProviderAffiliation, ProviderVisit, ProviderVisitChangedEvent, ProviderVisitDtIProductMsg, ProviderVisitMarketingItem, ProviderVisitProdDetailing, ProviderVisitProdDiscussion, ProviderVisitRqstSample, and PrvdVstSmplLmtTransaction objects.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

ProviderAcctProductInfo

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ProviderAcctProductInfo object.

Trigger Name	Description	Trigger condition
ProviderAcctProductInfoHandler	Sets the value of the Territory field on the provider account product information record to Null whenever a record is created or updated.	beforeInsert, beforeUpdate

ProviderAcctTerritoryInfo

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ProviderAcctTerritoryInfo object.

Trigger Name	Description	Trigger condition
AccountTerritoryFieldsHandler	Handles a valid territory check for an account and manages Provider Account Territory Information Sharing.	beforeInsert, afterInsert, beforeUpdate
ATFPreferredAddressHandler	Updates the Preferred Address field on the Provider Account Territory Information object from the account's Primary address in the Contact Point Address Object. This preferred address then displays in the Account Territory Fields component on the Account Dashboard.	beforeInsert, beforeUpdate
NewAlignmentConsentSharingHandler	Creates sharing records for Consent and Snapshot records based on territory assignments.	afterInsert, afterUpdate
PATIRejectionHandler	Deletes provider account territory info sharing and object territory association records when the status of the associated provider account territory info records changes to Rejected.	beforeUpdate

ProviderActivityGoal

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ProviderActivityGoal object.

Trigger Name	Description	Trigger condition
ProviderActivityGoalHandler	Validates and prevents the creation of duplicate Provider Activity Goals by ensuring the combination of Activity Plan and Account is unique. The trigger performs this validation when you create or update a Provider Activity Goal record. If a duplicate combination is found, an error is displayed, preventing the creation or update of the record.	beforeInsert, afterInsert, beforeUpdate, afterUpdate, beforeDelete, afterDelete, afterUndelete

ProviderAffiliation

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ProviderAffiliation object.

Trigger Name	Description	Trigger condition
AffiliationUniquenessGenericHandler	Ensures a unique combination of fields configured in the field set of the Provider Affiliation object, preventing duplicates and maintaining accurate records.	beforeInsert, beforeUpdate
AffiliationUniquenessHandler	Ensures affiliation uniqueness by enforcing a unique combination of account, related account, and role values. This prevents duplicate affiliations and maintains data integrity.	beforeInsert, beforeUpdate

Trigger Name	Description	Trigger condition
HardAffiliationHandler	Assigns the affiliation type of provider affiliation records as either hard or soft. Primary Affiliations are designated as Hard Affiliations. The HardAffiliationHandler trigger handler also ensures that duplicate Hard type affiliations between the same two accounts aren't created. Sets the affiliation type of workplace affiliations between an HCP and an HCO to Hard. It also sets the affiliation between an account and its related account to Hard when the related account is the preferred workspace of the account's provider account territory info (PATI) record. Lastly, it sets the affiliation type of reciprocal affiliations to None.	beforeInsert, afterInsert, beforeUpdate
AffiliationReciprocalHandler	Manages the creation, updating, and deletion of reciprocal affiliations. It ensures that the inverse affiliation is maintained whenever a ProviderAffiliation is created, updated, or deleted. Reverse Affiliations are only created if the Influence Type field on the Provider Affiliation object is selected as either Unidirectional, Bidirectional, or Neutral.	afterInsert, afterUpdate, beforeDelete
AffiliationPrimaryChangeHandler	Ensures a single primary	afterInsert, afterUpdate

Trigger Name	Description	Trigger condition
	affiliation between an account and a related healthcare organization account. When a new primary affiliation is created, the trigger deselects the primary flag on any prior affiliations with the same Account and Related Account. Additionally, it updates the associated Healthcare Provider's Primary Organization Account to the affiliation's Related Account, or creates a new Healthcare Provider record if one doesn't exist, populating it with the affiliation's Account and Related Account.	
AffiliationAccountUpdateHandler	Prevents the modification of the account and the related account after an affiliation is created.	beforeUpdate

ProviderVisit

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ProviderVisit object.

Trigger Name	Description	Trigger condition
PVUpdateChildFieldsHandler	Maintains data consistency for group visits by syncing child provider visit records with the parent provider visit records. Automatically updates key fields (Planned Visit Start Time, Planned Visit End Time, and Channel) when the parent visit is modified.	afterUpdate

Trigger Name	Description	Trigger condition
PVLockHandler	Prevents editing provider visit data based on the visit's status and signature state to maintain data integrity.	beforeUpdate, beforeDelete
RemoteSessionInvitationPVHandler	Manages remote session invitations by creating video calls, email notifications, and session keys for provider visit records.	afterInsert, afterUpdate, beforeDelete

ProviderVisitChangedEvent

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ProviderVisitChangedEvent object.

Trigger Name	Description	Trigger condition
VisitChangedEventShareHandler	Creates Visit shares with the users shared with Accounts.	afterInsert
PVChangedEventUpdateAccountTerrInfo	Processes provider visit changed events and updates provider visit related fields on Provider Account Territory Information records.	afterInsert

ProviderVisitDtIProductMsg

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ProviderVisitDtIProductMsg object.

Trigger Name	Description	Trigger condition
ProviderVisitDtIProductMsgLockHandler	Locks the ProviderVisitDtIProductMsg record when Visit is Submitted or Signed.	beforeInsert, beforeUpdate, beforeDelete

ProviderVisitMarketingItem

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the

ProviderVisitMarketingItem object.

Trigger Name	Description	Trigger condition
ProviderVisitMarketingItemLockHandler	Locks the ProviderVisitMarketingItem record when Visit is Submitted or Signed.	beforeInsert, beforeUpdate, beforeDelete

ProviderVisitProdDetailing

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ProviderVisitProdDetailing object.

Trigger Name	Description	Trigger condition
ProviderVisitProdDtlNameHandler	Generates and populates the Product Hierarchy Name field for the Provider Visit Product Detailing record.	beforeInsert, beforeDelete
ProviderVisitProdDtlLockHandler	Locks the ProviderVisitProdDetailing record when Visit is Submitted or Signed.	beforeInsert, beforeUpdate, beforeDelete

ProviderVisitProdDiscussion

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ProviderVisitProdDiscussion object.

Trigger Name	Description	Trigger condition
ProviderVisitProdDiscussionLockHandler	Locks the ProviderVisitProdDiscussion record when Visit is Submitted or Signed.	beforeInsert, beforeUpdate, beforeDelete

ProviderVisitRqstSample

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the ProviderVisitRqstSample object.

Trigger Name	Description	Trigger condition
ProviderVisitRqstSampleLockHandler	Locks the ProviderVisitRqstSample record when Visit is Submitted or Signed.	beforeInsert, beforeUpdate, beforeDelete
CalculateShippedQuantityHandler	Manages Provider Visit Request Sample shipped quantity on Visit.	afterUpdate

PrvdVstSmplLmtTransaction

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the PrvdVstSmplLmtTransaction object.

Trigger Name	Description	Trigger condition
SampleLimitTransactionHandler	Processes Sample Limit Records using Rule which comes from Web and Mobile as Provider Visit Sample Limit Transaction records.	afterInsert

Trigger Handlers for the SurveySubject - Visitor Objects

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the SurveySubject, Territory2, TerritoryAcctRcmdAction, TerritoryBusinessPlan, TerritoryContentTplAsgnt, TerritoryUserDowntime, Visit, and Visitor objects.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

SurveySubject

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the SurveySubject object.

Trigger Name	Description	Trigger condition
SurveyInvitationSharingHandler	Applies survey sharing logic on the Survey Subject object. When survey subject records are created or updated, this trigger handler automatically creates survey invitation share records based on one of three strategies: Account-Based Sharing: When a survey subject is linked to an account, the trigger handler automatically creates survey invitation share records for accounts listed in the related account's account share records. This sharing logic doesn't apply to records where the rowCause field is set to Owner. Life Sciences Marketable Product-Based Sharing: When you link a survey subject record to a Life Sciences marketable product, the corresponding survey invitation share records are created for either accounts or territories based on the product territory availability records. The availability of the product is determined by the product territory availability object, which includes an alignment type field with options such as territory subordinates, territory inclusion, or territory exclusion.	beforeInsert, afterInsert, afterUpdate

Trigger Name	Description	Trigger condition
	Territory-Based Sharing: When a survey subject record is linked to a territory, survey invitation share records are created for that territory and its subordinates.	

Territory2

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the Territory2 object.

Trigger Name	Description	Trigger condition
Territory2Handler	Prevents the deletion of a Territory2 record when related records like ProviderAcctTerritoryInfo, Inquiry, ContactPointBestContactTime, TerritoryAcctRcmdAction, TerritoryAccountScore, TerritoryAcctProdMsgScore, TerritoryUserDowntime, or TerritoryBusinessPlan records reference it.	beforeDelete

TerritoryAcctRcmdAction

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the TerritoryAcctRcmdAction object.

Trigger Name	Description	Trigger condition
TerrAcctRcmdActionSharingHandler	Shares the territory account recommended action records with the specified territory.	afterInsert, afterUpdate
TerrAccRcmActStatusUpdateHandler	Updates the status of the	afterInsert, afterUpdate

Trigger Name	Description	Trigger condition
	territory account recommended action when a visit is created.	

TerritoryBusinessPlan

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the TerritoryBusinessPlan object.

Trigger Name	Description	Trigger condition
TerritoryBusinessPlanDeleteHandler	Deletes Territory Business Plan records.	beforeDelete
TerritoryBusPlanChangeStatusHandler	Updates the statuses of Territory Business Plan records.	beforeInsert, beforeUpdate

TerritoryContentTplAsgnt

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the TerritoryContentTplAsgnt object.

Trigger Name	Description	Trigger condition
EmailTemplateTerritorySharingHandler	Updates territory sharing based on Territory Content Template Assignment records.	afterInsert, afterUpdate, afterDelete

TerritoryUserDowntime

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the TerritoryUserDowntime object.

Trigger Name	Description	Trigger condition
TUDUpdateHandler	Updates the start time and end time of time off territory records based on the time slot interval configuration.	beforeInsert, beforeUpdate
UDBHValidationHandler	Prevents the creation of time	beforeInsert, beforeUpdate

Trigger Name	Description	Trigger condition
	off territory records with start and end times outside of the org's business hours.	
TUDSharingHandler	Shares a user's time off territory records with other users in the same territory, giving them read-only access.	afterInsert
TUDVisitValidationHandler	Prevents the creation of time off territory during a visit.	beforeInsert, beforeUpdate
TUDHolidayValidationHandler	Prevents the creation of time off territory records on holidays.	beforeInsert, beforeUpdate
TUDOverlapHandler	Prevents the creation of overlapping time off territory records.	beforeInsert, beforeUpdate
TUDEventValidationHandler	Prevents the creation of time off territory during a general event.	beforeInsert, beforeUpdate

Visit

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the Visit object.

Trigger Name	Description	Trigger condition
VisitLockHandler	Prevents editing visit data based on the status and signature state to maintain data integrity.	beforeInsert, beforeUpdate
VisitTimeOffValidationHandler	Prevents scheduling a visit that conflicts with a time-off territory to support scheduling efficiency and accurate visit reporting.	beforeInsert, beforeUpdate
VisitLifeScienceAccountListLinkHandler	Clears the Life Science Account List field on the provider visit record if the	beforeUpdate

Trigger Name	Description	Trigger condition
	Planned Start Time is changed on a visit created by applying a routine.	
VisitHolidayValidationHandler	Prevents scheduling a visit on holidays or outside of business hours to maintain compliance with company policies.	beforeInsert, beforeUpdate
VisitPopulateEndTimeHandler	Calculates and sets the visit's Planned End Time by adding the default duration defined in the “Set blank date/time in Planned Start Time” Visit Administration setting in the Admin Console.	beforeInsert, beforeUpdate
FutureVisitLimitValidationHandler	Prevents scheduling visits beyond the limit specified in the “Future Visit Limit in Days” setting in Visit Administration in the Admin Console.	beforeInsert, beforeUpdate
VisitUpdateChildFieldsHandler	Maintains data consistency in group visits by synchronizing child visit records with their parent visits. Automatically updates key fields (Planned Visit Start Time, Planned Visit End Time, and Channel) when the parent visit is modified.	afterUpdate
VisitDelayReasonHandler	Clears the previously entered reason for delay if a user changes the visit date and time and the date no longer exceeds the submission delay threshold.	beforeUpdate
VisitCascadeDeleteHandler	Deletes attendee visits automatically when the parent group visit is deleted, maintaining data integrity.	beforeDelete

Trigger Name	Description	Trigger condition
VisitSurveyInviteCascadeDeleteHandler	Deletes survey invitations linked to a visit when a visit is deleted, and prevents deleting them if the visit is submitted to maintain data integrity.	beforeDelete
RemoteSessionInvitationVisitHandler	Manages remote session invitations by creating video calls, email notifications, and session keys for visit records.	afterUpdate, beforeDelete
VisitMyBestTimeValidationHandler	Prevents scheduling visits outside of best times defined for the account address and user's territory, improving scheduling efficiency and aligning with account availability.	beforeInsert, beforeUpdate
VisitChangePrimaryAccountHandler	Performs validations, updates related records, and prepopulates fields in the visit to maintain data consistency when the primary account is changed.	beforeInsert, beforeUpdate, afterUpdate

Visitor

Review the names, descriptions, and trigger conditions of the trigger handlers associated with the Visitor object.

Trigger Name	Description	Trigger condition
VisitorLockHandler	Prevents insert/update/delete operation on Visitor if it is locked.	beforeInsert, beforeUpdate, beforeDelete
VisitorShareHandler	Shares the Visit record with the Visitors (employees).	afterInsert, afterUpdate, afterDelete

Workflows for Life Sciences

Life Sciences Cloud for Customer Engagement workflows automate your business processes and guide

users through their tasks with clear steps. Configure the permissions and actions for each step of the workflow based on the users' context and role. Implement custom validations to make sure that users follow business processes, and provide checklists to help users understand how to use the workflow. These workflows are supported both online and offline in the Life Sciences Cloud mobile app.

Get Your Org Ready for Life Sciences Workflows

Before you create a Life Sciences workflow, complete these steps.

Understand Life Sciences Workflows

Life Sciences Cloud for Customer Engagement workflows outline the journey of an object through various stages and help you manage different statuses, actions, and record permissions. Get familiar with these terms, and understand how each part supports the overall workflow configuration.

Prepare to Configure a Life Sciences Workflow

Life Sciences Cloud for Customer Engagement workflows are a highly customizable way to streamline work for your field teams. To make sure your workflows function as expected, review these planning steps so you don't have to gather information or make decisions during setup.

Manage Life Sciences Workflows

To streamline your business processes, create tailored Life Sciences Cloud for Customer Engagement workflows.

Considerations for Life Sciences Workflows

When you configure a Life Sciences Cloud for Customer Engagement workflow, keep these considerations in mind.

Custom Scripts for Life Sciences

Custom scripts are programmatic tools for data validation across Life Sciences Cloud for Customer Engagement on desktop and both online and offline in the Life Sciences Cloud mobile app. Custom scripts are used in Life Sciences workflow management and visit management.

Get Your Org Ready for Life Sciences Workflows

Before you create a Life Sciences workflow, complete these steps.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

USER PERMISSIONS NEEDED

To access and work with Life Sciences Cloud for Customer Engagement data:	Life Sciences Commercial Admin permission set
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To update permission sets:	View Setup and Configuration
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1. Assign users a permission set that includes the Use Life Sciences Workflow Management permission.
2. To make sure that users can see and work with workflow configurations, [enable permissions for these objects](#) in the correct permission sets.
 - Life Science Custom Scripts
 - Life Science Stage Actions
 - Life Science Stage Objects
3. To make sure that users can access the workflow progress bar and data on the record detail page, [update the sharing settings](#) for these objects and set the default internal access to **Public Read Only**.
 - Life Science Custom Script
 - Life Science Stage Action
 - Life Science Stage Object

Alternatively, share these records with each user manually.

4. Activate the StagePathPermissionsHandler trigger handler for Life Sciences Customer Engagement workflows. See [Trigger Handler Administration](#).
5. Set up the configurations for the workflow to work in the Life Sciences Cloud mobile app.
 - a. [Set up object metadata cache configurations](#) for all workflow objects.
 - b. [Generate a metadata cache](#) to package the object schema configuration that the Life Sciences Cloud mobile app uses for online and offline access.

Understand Life Sciences Workflows

Life Sciences Cloud for Customer Engagement workflows outline the journey of an object through various stages and help you manage different statuses, actions, and record permissions. Get familiar with these terms, and understand how each part supports the overall workflow configuration.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

For an example workflow configuration, see [Set Up Workflow Actions for Medical Inquiry Management](#) and [Set Up the Workflow Stages to Manage Medical Inquiries](#).

Workflow Actions

Workflow actions are the actions that users can perform as part of the workflow.

Workflow Path

The workflow path is the object's overall workflow, and the path from one stage in the workflow to the

next.

Workflow Stage Value

Workflow stage values are the steps in the object's workflow path. A controlling picklist field determines the stages. For example, if the Status field is the controlling field on a workflow for the Inquiry object, the workflow path often includes stages such as New, Working, Escalated, and Closed.

Stage Object

The stage object is the object for which a workflow path is configured. When you create and save a workflow path, stage object records are created automatically. One stage object record is created for each object that has a workflow configured.

Stage Operation

Stage operations are a grouping of conditions that the record must meet in order to grant permissions and determine the actions that are available to users. You define stage operations for each stage in the workflow.

Stage Operation Conditions

Stage operation conditions are the specific conditions within the stage operation that a record must meet in order to grant permissions and make actions available to users. If the conditions aren't met, the permissions and actions aren't granted. You define stage operation conditions for each stage operation within each stage of the workflow.

Stage Operation Actions

Stage operation actions are the actions that are available to users when the stage operation is applied to a particular stage in the workflow path. You create workflow actions separately, and then you assign them as stage operation actions for each stage in the workflow's path.

Custom Scripts

Custom scripts are Lightning web components that validate user actions or provide checklists to help users understand the tasks to complete.

Prepare to Configure a Life Sciences Workflow

Life Sciences Cloud for Customer Engagement workflows are a highly customizable way to streamline work for your field teams. To make sure your workflows function as expected, review these planning steps so you don't have to gather information or make decisions during setup.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

Step 1: Outline the Workflow

Identify the business process that you want to automate. Then, break down the workflow into logical steps or stages. For example, a common Inquiry process includes stages such as New, Working, Escalated, and Closed.

Step 2: Define Workflow Objects and Permissions

Identify the parent object that you want the workflow to run on, as well as any related objects. Determine the permissions users need to create, edit, or delete parent or child records at each stage in the object's workflow.

Step 3: Identify Fields for Workflow Conditions

Each workflow stage operation includes conditions that must be met so that users can work with records and take actions. Identify the fields on the parent object or related objects that are required to set up these conditions.

Step 4: Plan Actions for Each Workflow Stage

Workflow actions can update records, launch a platform event, invoke a Lightning web component, or open a custom URL. Outline the actions that you want to make available to users at each stage in the workflow. This way, you can identify the record fields or other components that the actions use or update.

Step 5: Consider Validation Rules and Checklists

Validation rules and checklists help you maintain data integrity and enforce business rules. For validation rules, consider what conditions must be met for a record to move to the next stage in the workflow. For checklists, identify any necessary steps that you want to outline for users so that they know how to move a record to the next stage.

Manage Life Sciences Workflows

To streamline your business processes, create tailored Life Sciences Cloud for Customer Engagement

workflows.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

You can manage Life Sciences workflows from the Workflow Configuration tile in the Admin Console. From the Workflow Paths page, you can:

- Search the list of workflow paths, and sort by column to find the workflow you need.
- Get details about each workflow path, such as the object and record type it applies to, its status, and its controlling field.
- Create a workflow path.
- Clone a workflow path.
- Open or edit a workflow path to modify its details, activate it, or deactivate it.

After you configure a Life Sciences workflow, you can use Data Loader to move the workflow configuration to another org.

Configure Life Sciences Workflow Actions

Before you configure a workflow path, create the actions that users can perform on an object. When you define the workflow path, you can assign these actions at each stage in the workflow.

Configure Life Sciences Workflow Paths

Create workflow paths for each object and record type to meet your business needs. At each stage of the record's lifecycle, define user permissions and the actions that users can take.

Configure Custom Scripts for Life Sciences Workflows

To make sure that records meet your business rules before the Life Sciences Customer Engagement workflow continues, create and manage custom scripts. Each time a user performs an action, validation scripts run to prevent incorrect actions or alert users about errors. Checklist scripts show an info icon on Update Record actions so that users can see the next steps to take.

Assign Custom Scripts to Life Sciences Workflow Objects

To enforce business rules throughout the workflow, update the validation scripts and checklists that apply to each stage object.

See Also

[Data Loader Guide](#)

Configure Life Sciences Workflow Actions

Before you configure a workflow path, create the actions that users can perform on an object. When you define the workflow path, you can assign these actions at each stage in the workflow.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

USER PERMISSIONS NEEDED

To create workflow actions:	Life Sciences Commercial Admin permission set
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You can create four types of workflow actions.

- Update Record actions edit records in some way, for example, by moving the record to a new status.
- Platform Event actions trigger real-time processes. For example, a platform event can launch a flow.
- Component actions invoke a Lightning web component. For example, you can open a component that shows a list of available document templates.
- Custom Actions support custom functionality, such as opening an external URL. You create custom actions from the Quick and Custom Action Administration page in the Admin Console. Actions must have the Stage Path entity type and the URL action type. See [Create Custom Actions](#).

1. From the App Launcher, find and select **Admin Console**.
2. Select **Workflow Configuration**, and then select **Workflow Actions**.
To see and search for available actions, select the tabs for each action type.
3. To create an action, click **New**, and select the action type that you want to create.
4. Enter the action's basic information, such as its name, the button label, and the object on which it's available.
5. For an Update Record action, enter these details.
 - a. Select the field on this object that the action updates, and enter the field's default value.
 - b. In the Target Object section, select an additional object, add the fields to update, and enter the default field values.
When this action is triggered, a window opens where users can review, update, and save changes to these fields.
 - c. In the Platform Event Parameters section, select a platform event to publish as part of this action, add the fields to include, and specify the default values.
The platform event details are sent as parameters to workflows, Apex processes, or other asynchronous processes that are subscribed to this platform event.
6. For a Platform Event action, enter these details.
 - a. In the Target Object section, select an additional object, add the fields to update, and enter the default field values.
When this action is triggered, a window opens where users can review, update, and save changes to these fields.
 - b. In the Platform Event Parameters section, select a platform event to publish as part of this action, add the fields to include, and specify the default values.
The platform event details are sent as parameters to workflows, Apex processes, or other

asynchronous processes that are subscribed to this platform event.

7. For a Component action, enter these details.
 - a. In the Component Parameters section, enter the name of the Lightning web component to open from this action.
 - b. To populate field values when the component loads, enter component parameters.
For example, for an action that opens a component to capture a signature, add these parameters to set these values when the component loads:

```
"status": "Signed", "disclaimerText": "Disclaimer Text".
```
8. For a Custom Action, select a custom URL action to open an external link.
9. Save your changes.

After you create actions, you can see action details and parameters on the Workflow Actions page. You can also edit, clone, or delete actions.

See Also

[Define and Manage Platform Events](#)

[Lightning Web Components Developer Guide: Create Lightning Web Components](#)

Configure Life Sciences Workflow Paths

Create workflow paths for each object and record type to meet your business needs. At each stage of the record's lifecycle, define user permissions and the actions that users can take.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

USER PERMISSIONS NEEDED

To configure workflow paths:	Life Sciences Commercial Admin permission set
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1. From the App Launcher, find and select **Admin Console**.
2. Select **Workflow Configuration**, and then select **Workflow Paths**.
3. Click **New**, and give the workflow path a name.
4. Select the object and record type to create the workflow path for.
5. To determine the stages in the workflow path, select the controlling picklist field.
6. To open the workflow builder and define the workflow path, click **Continue**.
The workflow builder opens on the first stage of the workflow. To select a different stage, click its name in the progress bar at the top of the page.
7. Define a stage operation, which groups the conditions that control the permissions that users have, and the actions they can take when records are in this stage.
 - a. Under Stage Operations, click the plus icon.

- b. Name the operation.
 - c. Select whether users can create, edit, or delete records at this stage when conditions are met.
Standard Salesforce object and field-level permissions still apply.
 - d. Enter a priority for this operation.
When multiple stage operations apply to a record, the highest priority operation takes precedence, and users are granted those permissions.
 - e. Save your changes.
8. Define stage operation conditions that must be met in order to grant users permissions and show configured actions for this stage and operation.
 - a. Select **Stage Operation Conditions**, and then click the plus icon.
 - b. Select the field to evaluate.
You can select fields on this object or a related object.
 - c. Set the operator for the condition, and enter the required field value.
 - d. Add more conditions as needed.
All conditions must be met to grant permissions and actions.
9. Define the stage operation actions, which determine the actions that users can perform when conditions are met for this stage.
 - a. Select **Stage Operation Actions**.
 - b. Click **Add Actions**, and then select the type of action to add.
 - c. Select the actions to add to the page, and then save your changes.
 - d. Add more actions as needed. To reorder the actions, use the arrows.
10. In the Other Permissions tab, choose whether to grant users access to Salesforce Files.
With this permission, users can manage documents and attachments at this stage when conditions are met.
11. Add or edit additional stage operations for this stage, if needed.
 - a. To add another parent stage operation, click the plus icon next to Stage Operations.
 - b. To add a child stage operation, click the plus icon next to a stage's name.
Child stage operations control access to child objects of the workflow path's object.
For example, in an Inquiry workflow path, create a child stage operation to control when users can create, edit, or delete Inquiry Question records.
 - c. To edit a stage operation, select it under Stage Operations, and then click the pencil icon.
 - d. To delete a stage operation, select it under Stage Operations, and then click the trash can icon.
 - e. To clone a stage operation and make minor changes, select it under Stage Operations, and then click the copy icon.
12. Similarly, configure at least one stage operation for each stage in the workflow.
When a stage is configured, a green checkmark appears in the progress bar.
13. After your workflow is completed, activate it to make it available to that object or object type, or save your changes and activate it later.
14. To create a copy of this workflow path, click **Clone**.

See Also

[Considerations for Life Sciences Workflows](#)

Configure Custom Scripts for Life Sciences Workflows

To make sure that records meet your business rules before the Life Sciences Customer Engagement workflow continues, create and manage custom scripts. Each time a user performs an action, validation scripts run to prevent incorrect actions or alert users about errors. Checklist scripts show an info icon on Update Record actions so that users can see the next steps to take.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

USER PERMISSIONS NEEDED

To configure custom scripts:	Life Sciences Commercial Admin permission set
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First, configure custom scripts for the validations that you want to perform or for the checklists that you want to apply. See [Create Lightning Web Components](#) and [Custom Scripts for Life Sciences](#).

Then, you can apply the custom scripts from the Admin Console in Life Sciences Customer Engagement.

1. From the App Launcher, find and select **Admin Console**.
2. Select **Workflow Configuration**, and then select **Custom Scripts**.
3. Click **New**, and give the script a name.
4. Enter the name of the Lightning web component that you created.
5. Select whether this is a script for a checklist or for validation.
6. Save your changes.
7. To sync changes to the customer script after you update the Lightning web component, select **Refresh**.

After you configure custom scripts, assign them to workflow objects.

Assign Custom Scripts to Life Sciences Workflow Objects

To enforce business rules throughout the workflow, update the validation scripts and checklists that apply to each stage object.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

USER PERMISSIONS NEEDED

To assign custom scripts to workflow objects:	Life Sciences Commercial Admin permission set
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Stage object records are created automatically after you create and save a Life Sciences Cloud for Customer Engagement workflow path. One stage object record is created for each object that has a workflow. Validations and checklists apply to all object types.

1. From the App Launcher, find and select **Admin Console**.
2. Select **Workflow Configuration**, and then select **Stage Objects**.
3. Find the object to apply custom scripts to, and then select **Edit** from the row-level actions.
4. In the Validation Script field, select the custom script that you created for validation.
5. In the Checklist Script field, select the custom script that you created for the checklist items.
6. Save your changes.
After you save your changes, custom scripts run automatically as long as the object has an active workflow.

See Also

[Configure Custom Scripts for Life Sciences Workflows](#)
[Custom Scripts for Life Sciences](#)

Considerations for Life Sciences Workflows

When you configure a Life Sciences Cloud for Customer Engagement workflow, keep these considerations in mind.

REQUIRED EDITIONS

Available in: Lightning Experience

Available in: **Enterprise** and **Unlimited** Editions with Life Sciences Cloud, Life Sciences Cloud for Customer Engagement Add-on license, and the Life Sciences Customer Engagement managed package.

Conflicting Workflow Paths for Parent and Child Objects

Two workflow paths can control permissions to an object record at the same time. This happens if a parent object's workflow path applies a child stage operation for the child object, and the child object has its own workflow path and stage operation. When a record's permissions are controlled by two separate workflow paths, the most restrictive permissions apply.

For example, let's say you have separate workflow paths for the Inquiry object and the child Inquiry Question object. The Inquiry Question workflow path includes a stage operation that grants users only the Delete permission when the response status is Responded. The Inquiry workflow path includes a child stage operation that grants users Create, Edit, and Delete permissions on Inquiry Question records when the response status is Responded or Not Responded. Because the Inquiry Question workflow path

is more restrictive, users can't create or edit Inquiry Question records with a Responded status.

Workflow Paths for Objects with Record Types

If an object has record types, you must configure a workflow path for every record type. If you create workflow paths only for some record types, users can't create, edit, or delete records of other types.

Workflow Path Stages

Each workflow stage requires at least one stage operation. If a stage doesn't have a stage operations:

- Users can't create, update, or delete records when the record is at that stage.
- The stage doesn't appear in the progress bar at the top of the record detail page.

Workflow Path Field Lengths

Text fields in the workflow path support a maximum length of up to 255 characters. With this length limit, you can reference parent objects in the workflow path, for example, in stage operation conditions.

Stage Operation Priority

When a workflow stage has multiple operations, a record in that stage can meet the conditions for more than one operation. To make sure the correct operation is applied and users are granted the correct permissions, assign each operation a priority.

For example, the New stage in the Inquiry workflow path has two operations:

- Inquiry New Stage Operation 1" has condition "City==San Francisco AND State==CA"
- Inquiry New Stage Operation 2" has condition "State==CA"

A new Inquiry record in San Francisco, California meets both conditions. Assign the higher priority to Operation 1 to grant users those permissions.

Workflow Record Update Actions

When a user runs a Record Update action, the action executes in this order.

1. The validation script runs.
2. The window opens where users can review, update, and save changes to additional object fields. For example, a window opens where a user can enter additional information about why they're canceling an inquiry.
3. The primary record is updated.
4. The platform event fires.